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Toward intelligent tunnel construction: The universal discontinuity index for rapid probabilistic prediction of progressive batch rock-block failure—A theoretical, numerical, and experimental validation framework

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ABSTRACT

The safe, cost-effective advance of deep tunnels increasingly hinges on real-time forecasts of structurally controlled rock-block failure—capabilities that lie beyond classic Key-Block Theory (KBT) and remain computationally prohibitive for full-physics numerical models. We introduce the Universal Discontinuity Index (UDi), a single probabilistic score that fuses four physically grounded components—physical damage, effective-stress intensity, active-fracture ratio, and kinematic feasibility—into a scalable indicator of block instability. The index embodies a complete-system view: it treats the fracture network, local stress field, and block release kinematics as an integrated, dynamically updating subsystem within a Digital-Twin (DT) workflow.

Three levels of validation are presented. (i) Theoretical equivalence: UDi's kinematic term reproduces Warburton's vector criteria while providing a continuous risk scale. (ii) Numerical benchmarks: in controlled wedge-formation tests, UDi captured progressive batch failure with the same temporal hierarchy recorded by Bonded-Particle DEM and hybrid FDEM, yet required milliseconds rather than hours. (iii) Field proof: 52 photogrammetry-mapped faces from Chile's El Teniente mine were analysed via a stochastic DFN–KBT engine (>5 300 realizations); the normalized UDi predicted the mean number of unstable blocks with $R2 = 0.93$ and 95 % accuracy.

Leveraging its speed, we embed UDi in a Poisson-regression early-warning model that converts each index update into an expected block count per advance round, validated via jack-knife resampling. Fracture-propagation-informed refinement extends the method to blocks still in formation, enabling continuous hazard maps as excavation proceeds.

UDi therefore bridges the gap between deterministic kinematic checks and high-fidelity simulations, delivering DT-ready, probabilistic, and progressive failure forecasts that unlock proactive support design and truly adaptive tunnel operations.

Abbreviations: **BPDEM**, Bonded Particle Discrete Element Method; **BEM**, Boundary Element Method; **DT**, Digital Twin; **DTMs**, Digital Terrain Models; **DEM**, Discrete Element Method; **DFN**, Discrete Fracture Network; **DDA**, Discontinuous Deformation Analysis; **FEM**, Finite Element Method; **FDM**, Finite Difference Method; **FDEM**, Finite Discrete Element method; **KBT**, Key Block Theory; **NMM**, Numerical Manifold Method; **UDi**, Universal Discontinuity Index.

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Nomenclature

a	constant coefficient in [0,1]
α	is the acute angle between the scanline and the normal vector of the discontinuity,
$C.V_{tot-ver}$	total circular variance with respect to the vertical direction
δ_i	Set of fracture diameter
ESIF	Effective Stress Intensity Factor
$\widehat{\varphi}(\delta_i)$	estimating function of values of fracture diameter
$\widehat{f}_1$	steepest direction on the fixed face
$f(l)$)	probability density function
$g(\delta)$	probability density function
$G(\delta)$	cumulative density function
Kine	measure the contribution of flaw into kinematic damage severity
$\widehat{m}$	a unit vector representing the direction of the block movement,
$\widehat{n_1}$	the unit normal of the fixed face 1 directed out of the face,
N_{AF}	Number of shear active fractures according to adapted failure criteria
N_{TF}	Total number of fractures within rock mass
N	denotes the number of fractures intersecting the scanline.
P32	volumetric fracture intensity
PD	Physical damage
$\widehat{R}$	the block's weight acting vertically downward on its center of mass.
μ	stands for the mean fracture diameter
w_1, w_2, and w_3	Weighting factors depending on rock engineering design and purposes.

1. Introduction

Underground tunneling is a fundamental yet highly expensive and delay-prone activity in modern mining and civil engineering projects. Large-scale operations such as the El Teniente mine in Chile, which spans ≈ 3,000 km of tunnels, exemplify the logistical and economic challenges involved. Studies have shown that the costs associated with underground excavation and tunnel failures can represent between 10 % and 30 % of the total production costs in major projects (Avendo Castillo et al., 2008). Moreover, underground construction projects are more prone to delays and cost overruns compared with surface construction . A significant source of risk and cost in tunneling arises from structurally controlled failures, where rigid rock blocks detach along pre-existing discontinuities. Characterizing these discontinuities accurately is particularly challenging owing to the inherent danger and time constraints of geotechnical surveys in active headings (Hazrathosseini & Mahdevari, 2019; Lu et al., 2022; Hekmatnejad et al., 2021a, Hekmatnejad et al., 2021b). In addition, the presence of epistemic uncertainties in fracture geometry significantly affects the reliability of stability predictions (Spross et al., 2018; Spross et al., 2020). This situation underscores the need for robust, quantitative, and rapid tools that can dynamically evaluate instability potential as new geological data are collected.

Current approaches to rock-mass stability assessment encompass direct mechanical testing on core or large in-situ samples, empirical classification systems, and numerical modeling techniques. While direct testing provides accurate results, it is impractical for continuous assessment during tunnel advance (Li et al., 2014). Empirical systems such as the Rock Mass Rating (RMR) and the Q-system (Bieniawski, 1978; Barton, 2002) are favored for their simplicity and speed, yet they remain qualitative, heavily operator-dependent, and predominantly based on empirical correlations rather than mechanistic principles. Analytical and numerical models, including Key Block Theory (KBT) (Warburton, 1981; Goodman & Shi, 1985), Discontinuous Deformation Analysis (DDA), the Discrete Element Method (DEM), the Finite Element Method (FEM), the Finite Difference Method (FDM), the Boundary Element Method (BEM), and hybrid Finite-Discrete Element Method (FDEM) (Deb & Das, 2010; Yang et al., 2018; Kang et al., 2022a), have become increasingly sophisticated and capable of simulating complex geomechanical behaviors. However, these methods often require expert-driven parameterization, complex geometric modeling, and significant computational resources, which limits their practicality for real-time application.

Recent advances in artificial-intelligence (AI)–assisted rock-slope analysis further highlight the move toward automated, data-driven workflows. For example, Zhu et al. (2022) proposed a deep-learning framework that classifies key, trapped, and stable blocks directly from slope imagery, achieving ≈ 95 % accuracy. Similarly, Azarafza et al. (2020a) introduced a fuzzy-logic decision-making algorithm that rapidly assigns a "degree of reliability" to wedge and planar blocks without exhaustive parameter input. These developments demonstrate that AI techniques can supplement traditional KBT by delivering near-instantaneous classifications and probabilistic scores.

Key Block Theory remains widely used because of its geometric simplicity and low computational cost (Kulatilake et al., 2011). Nevertheless, KBT's deterministic nature limits its capacity to model progressive or batch failures, manage large fracture datasets, or integrate efficiently with automated data pipelines (Cai et al., 2022). A comprehensive review of limit-equilibrium formulations for discontinuous slopes underscores these shortcomings and calls for hybrid approaches that couple KBT with probabilistic inference and numerical verification (Azarafza et al., 2017; Azarafza et al., 2021). Beyond classic wedge and planar kinematics, engineers must also contend with complex toppling scenarios. Azarafza et al. (2020b) developed a Python-based algorithm that unifies block-flexural-toppling mechanics within a semi-distinct-element framework, offering higher processing speed than full DEM while preserving block-level fidelity. Most recently, Mao et al. (2025) presented the BTSTSA method, an extended block-theory formulation that yields conservative factors of safety for secondary toppling and has been validated against the Aydan-Kawamoto benchmark.

The emergence of Digital-Twin (DT) technologies—dynamic, bidirectional digital replicas of physical assets—presents an opportunity to enhance geomechanical risk assessment. DTs have already transformed manufacturing (Rosen et al., 2015) and are increasingly applied in energy infrastructure (Wang et al., 2021). In tunneling, prototype DT frameworks incorporate photogrammetry, LiDAR, and microseismic data to enhance spatial awareness and monitoring (Naderi and Shojaei 2023; Ninić et al., 2024). However, a critical gap remains: the lack of a risk-analysis engine capable of integrating diverse data streams, quantifying damage probabilistically, and updating swiftly enough to guide operational decisions (Tuhaise et al., 2023).

To address this gap, this study introduces the Universal Discontinuity Index (UDi), a fully quantitative, probabilistic index that consolidates four physically grounded components—Physical Damage (PD), Effective Stress Intensity Factor (ESIF), the ratio of Active to Total Fractures (NAF/NTF) and a Kinematic Factor (Kine)—into a single instability measure. This composite index provides a scalable, real-time and computationally efficient assessment of rock-mass instability, tailored for integration within DT workflows and aimed at reducing the subjectivity inherent in empirical ratings (Hekmatnejad et al., 2022; Hekmatnejad et al., 2025).

This paper formulates the UDi framework for probabilistic

assessment of progressive batch rock-block failure, validating its performance against Key Block Theory and two advanced numerical schemes, namely the Bonded-Particle Discrete Element Method (BPDEM) and the Finite-Discrete Element Method (FDEM). Field validation is conducted on 52 photogrammetry-mapped tunnel sections from the El Teniente mine, where UDi demonstrated a 95 % predictive agreement with a hybrid DFN–Warburton model. Additionally, the study develops an early-warning predictive model using Poisson regression, translating UDi values into real-time forecasts of the number of unstable blocks per tunnel round. The results establish UDi as a fast, objective and DT-ready alternative to existing geomechanical assessment methods, combining predictive reliability with operational efficiency suitable for modern tunneling challenges.

2. The universal discontinuity index (UDi): A probabilistic semantic framework for damage quantification in fractured rock masses

The Universal Discontinuity Index (UDi) is presented as quantitative semantic framework for evaluating damage severity in fractured rock material. Functionally analogous to the damage variable (D) in continuum damage mechanics, UDi quantifies internal degradation by integrating the interactions between pre-existing discontinuities and in-situ stress fields (Hekmatnejad et al., 2022; 2025). The index is composed of four core components—Physical Damage (PD), Effective Stress Intensity Factor (ESIF), the ratio of shear-active to total fractures (NAF/NTF), and a Kinematic Factor (Kine)—each grounded in well-established principles from fracture and damage mechanics. These elements are synthesized into a single expression (See equation (1):

$$UDi = \mathrm{PD} \times \left[1 + w_1(\mathrm{ESIF}) + w_2\left(N_{AF}/N_{TF}\right) + w_3(Kine)\right] \quad (1)$$

Equation (1) illustrates the general formulation of UDi system and its components.

UDi: Universal Discontinuity Index.

PD: Physical damage.

ESIF: Effective Stress Intensity Factor.

N_{AF}: Number of shear active fractures according to adapted failure criteria.

N_{TF}: Total number of fractures within rock mass.

Kine.: measure the contribution of flaw into kinematic damage severity.

w_1, w_2, and w_3: are weighting factors reflecting the specific demands of the engineering design scenario.

2.1. Physical damage (PD)

PD accounts for the inherent structural degradation within the rock mass and is computed as the product of volumetric fracture intensity (P32), fracture size factor, and orientation entropy. The P32 parameter, which represents the cumulative area of fractures per unit rock volume, is central to PD and can be estimated reliably through linear surveys regardless of the surface morphology (Hekmatnejad et al., 2017). The value of P32 aligns with the concept of fracture energy and can be correlated with damage accumulation and seismicity trends during multi-event failure processes. Additionally, PD captures the scale-independent geometric likelihood of block formation, thus enabling a physically grounded damage representation that evolves dynamically with new fracture data (Hekmatnejad et al., 2021b).

2.2. Effective stress intensity factor (ESIF)

ESIF quantifies the geometric alignment between pre-existing flaws and the surrounding stress field. It is derived from the reduction in stress intensity at the flaw tip due to confinement, as defined by:

$$K_{I,eff} = K_I - K_{I,conf} K_{II,eff} = K_{II} - K_{II,conf}$$

These components reflect the differences between the applied and confined stress states for Mode I and II fractures. A higher ESIF implies a flaw is more aligned with failure under the prevailing stress conditions, thereby contributing more significantly to material degradation. The accurate calculation of ESIF requires knowledge of the stress regime and flaw characteristics, including orientation and fill material. Studies have shown that ESIF correlates strongly with the reduction in strength due to crack propagation mechanisms (Miao et al., 2018; Lemon, 2016, 2021).

2.3. Active fracture ratio (AFR)

The AFR component captures the proportion of fractures that are actively failing under the current stress conditions. The activation is assessed via classical failure criteria such as Mohr-Coulomb for shear (Mode II) and Griffith for tensile failure (Mode I), or via a critical stress intensity approach for mixed-mode failure.

This term facilitates a transition-based damage classification: when AFR = 0, the rock mass is in the elastic stage; when AFR > 0, it progresses to stable or unstable crack growth. This indicator allows differentiation between pre-failure and coalescence stages and has been validated through experimental observations showing that the ratio of Mode II to Mode I activity increases as failure approaches (Mahabadi, 2012; Miao, et al., 2023).

2.4. Kinematic factor (Kine)

Kine evaluates the likelihood of post-peak block movement and quantifies the contribution of individual or intersecting fractures to kinematic instability. It reflects the ability of blocks formed by intersecting discontinuities to mobilize under gravity or excavation-induced unloading. This factor is derived from a vector-based assessment consistent with Warburton's criteria, allowing continuous quantification of the release tendency of fractured blocks. UDi's kinematic term enables a probabilistic interpretation of post-peak behavior and differentiates between block formation and actual failure, which depends on directional feasibility. Experimental and numerical validation confirms that this term effectively distinguishes between stable and unstable block configurations across both lab and tunnel-scale scenarios.

The novelty of UDi lies in its complete system perspective and deep model integration, both conceptually aligned with philosophies from enterprise security system design (Cava, 2013; Hekmatnejad et al., 2025). Just as integrated security platforms unify subsystems—alarms, access controls, and surveillance—into a scalable and interoperable whole, UDi treats the rock mass as a dynamic system whose structural response depends on the configuration, relevance, and interaction of flaws under varying stress regimes. This system-level modeling approach enables UDi to adjust for the contextual importance of discontinuities, recognizing that features critical to stability in one engineering application (e.g., tunneling) may be negligible in another (e.g., slope design or fluid flow). Importantly, UDi supports cross-domain interoperability by offering a unified, semantically meaningful damage metric interpretable across disciplines, thereby facilitating more coherent communication and decision-making throughout the various engineering units. In its construction, UDi integrates a topological model of the discontinuity network, vector-based kinematic analysis using Warburton's criteria, Effective stress intensity factor, and the Mohr-Coulomb, Griffith theory framework for assessing shear failure and tension failure correspondingly.

The reliability and practical relevance of the Universal Discontinuity Index (UDi) have now been confirmed across every scale of interest to rock-engineering practice. Laboratory tests on gypsum and sandstone show that UDi tracks the combined influence of flaw orientation, specimen size, in-fill material, confinement pressure and post-peak block

kinematics, mirroring measured strength trends and softening behavior with high fidelity. Hybrid DFN–FDEM pillar simulations extend this validation to the mine-pillar scale: across 21 fracture geometries, peak strength falls systematically as UDi rises, whereas conventional GSI, RMR, Q and RMi fail to capture the same fracture-length and orientation effects. Its application in tunnel scale includes damage assessment (Hekmatnejad et al., 2022), spatial modeling of rock strength heterogeneity (Hekmatnejad et al., 2023), and quantitative assessment of structural controls on drill-and-blast overbreak (Cortés et al., 2025).

In addition, UDi can be automatically computed using the R-Dis-Frag computer package (Hekmatnejad and Benoit, 2022), which processes imported fracture attributes—such as size, location, orientation, and in-situ stress regime—within fractions of a second, thereby eliminating user dependency and ensuring consistency. Collectively, these studies validate UDi's robustness across both laboratory and field scales and highlight its utility in enhancing predictive performance and automation within digital twin frameworks for geotechnical applications.

3. Methodology

As illustrated in Fig. 1, the mere formation of blocks does not inherently imply instability; rather, it is the kinematic compatibility among fracture sets that governs the likelihood of block failure. In this study we choose the Design-objective tuning strategy. Because UDi is used in different engineering contexts, the weights reflect the purpose of the analysis. For example, to isolate kinematic feasibility in a key-block study, the weightings set w1 = 0, w2 = 0, w3 = 1. This approach is implemented in the present study.

This configuration enables a targeted probabilistic evaluation of how the geometric and kinematic arrangement of intersecting fractures contributes to block formation and potential instability, as demonstrated throughout the study.

Importantly, the UDi framework retains the flexibility to incorporate all relevant components—physical damage, fracture activity, and kinematics—when applicable. The weighting coefficients (w_1, w_2, w_3) can be empirically calibrated using Pearson correlation analyses between each component and field-observed failure indicators, followed by normalization (Cortés et al., 2025). This modularity allows UDi to be tailored to specific engineering contexts, available datasets, and design requirements.

In the subsequent sections, we evaluate the applicability and validation of the Universal Discontinuity Index (UDi) as a novel framework for probabilistic prediction of progressive batch rock block failure. This is achieved through a series of integrated theoretical, experimental, and numerical scenarios, as summarized in the Table 1.

The following flowchart (Fig. 2) illustrates the methodology designed to validate the Universal Discontinuity Index (UDi) through two main objectives, each targeting a different validation dimension: real-time probabilistic analysis of progressive block movement and real-scale identification of unstable blocks. This structured approach reflects the scenarios summarized in the previous table and ensures comprehensive evaluation of the UDi system across spatial and temporal domains of rock block failure.

Objective 1: Real-Time Probabilistic Progressive Analysis of Block Movement.

This pathway evaluates the UDi's performance in detecting the potential for batch rock block failures and its sensitivity to the formation of unstable blocks driven by the evolution of the fracture system:

Step #1: Conceptual Experiment DesignThree progressive failure experiments are conceptualized:

- o Experiment 1 simulate batch block failure evolution over time (tunnel collaps).
- o Experiment 2 & 3 simulate block geometry evolution over time (wedge formation at tunnel crown)

Step #2: Simulation-Based Kinematic AnalysisThe experiments are executed using:

- o The UDi system for probabilistic damage modeling,
- o The KBT approach for deterministic key block evaluation, and
- o A numerical model (e.g., Bonded Particle Discrete Element Method) for dynamic interaction analysis.

Objective 2: Real-Scale Identification of Unstable Block zones.

This pathway demonstrates UDi's capacity to detect potentially unstable blocks using real field data. It integrates fracture geometry, photogrammetry-based modeling, and comparative validation against

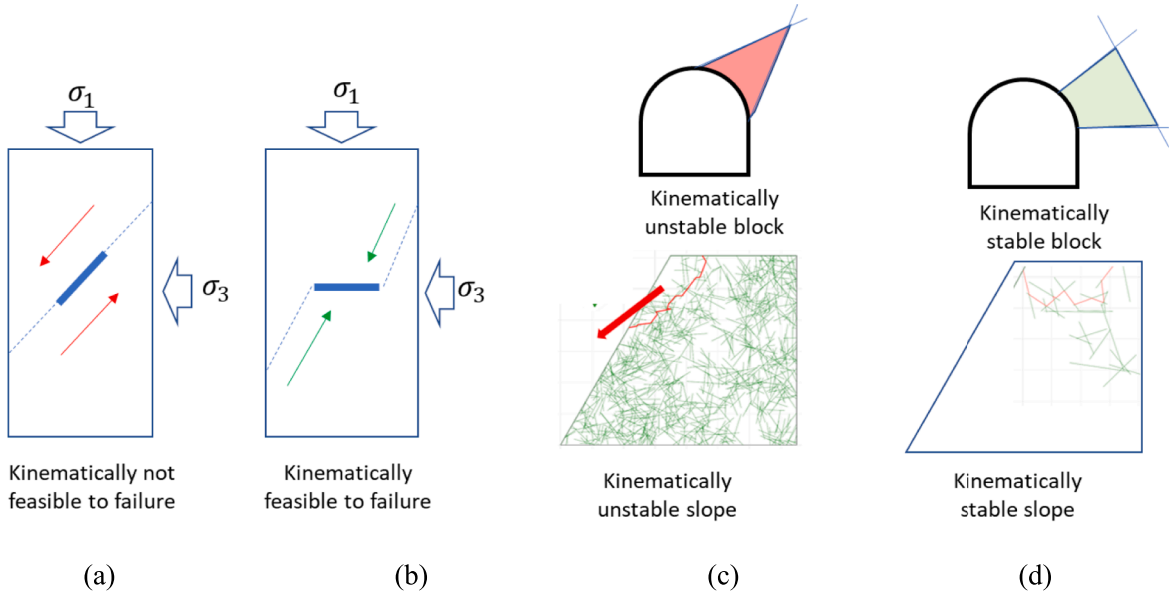


Fig. 1. Assessment of the single fracture and fracture system's influence on kinematic rock block failure at experiment at Lab scale (a and b) and tunnel and slope (c and d); a) pre-existing flaw is favorable for kinematic movement of the rock fragments, b) pre-existing flaw is not favorable for kinematic movement of the rock fragments, c) formation of the kinetically stable rock fragments in case of tunnel and slope, d) formation of the kinetically unstable rock fragments in case of tunnel and slope (Modified after Hekmatnejad et al., 2025).

Table 1
The proposed scenarios are specifically developed to illustrate the applicability and validate the effectiveness of the UDi framework in assessing progressive batch rock block instability.

Scenario	Focus	Validation Method	Comparison Baselines
Controlled Block Movement	Kinematic block instability	$UDivs.\{KBT$	θ angle-based classification
Progressive Batch Failure Simulation	Sequential batch block movement	$UDivs.\begin{cases} KBT \\ BPDEM \\ FDEM \end{cases}$	Dynamic validation over time
Fracture Propagation	Damage sensitivity to geometry	$UDivs.\begin{cases} KBT \\ BPDEM \\ FDEM \end{cases}$	Stepwise fracture evolution
Hazard Zonation and Predictive Model	Real-scale hazard mapping	$UDivs.\{$Hybrid probabilistic DFN + KBT	Spatial comparison with KBT

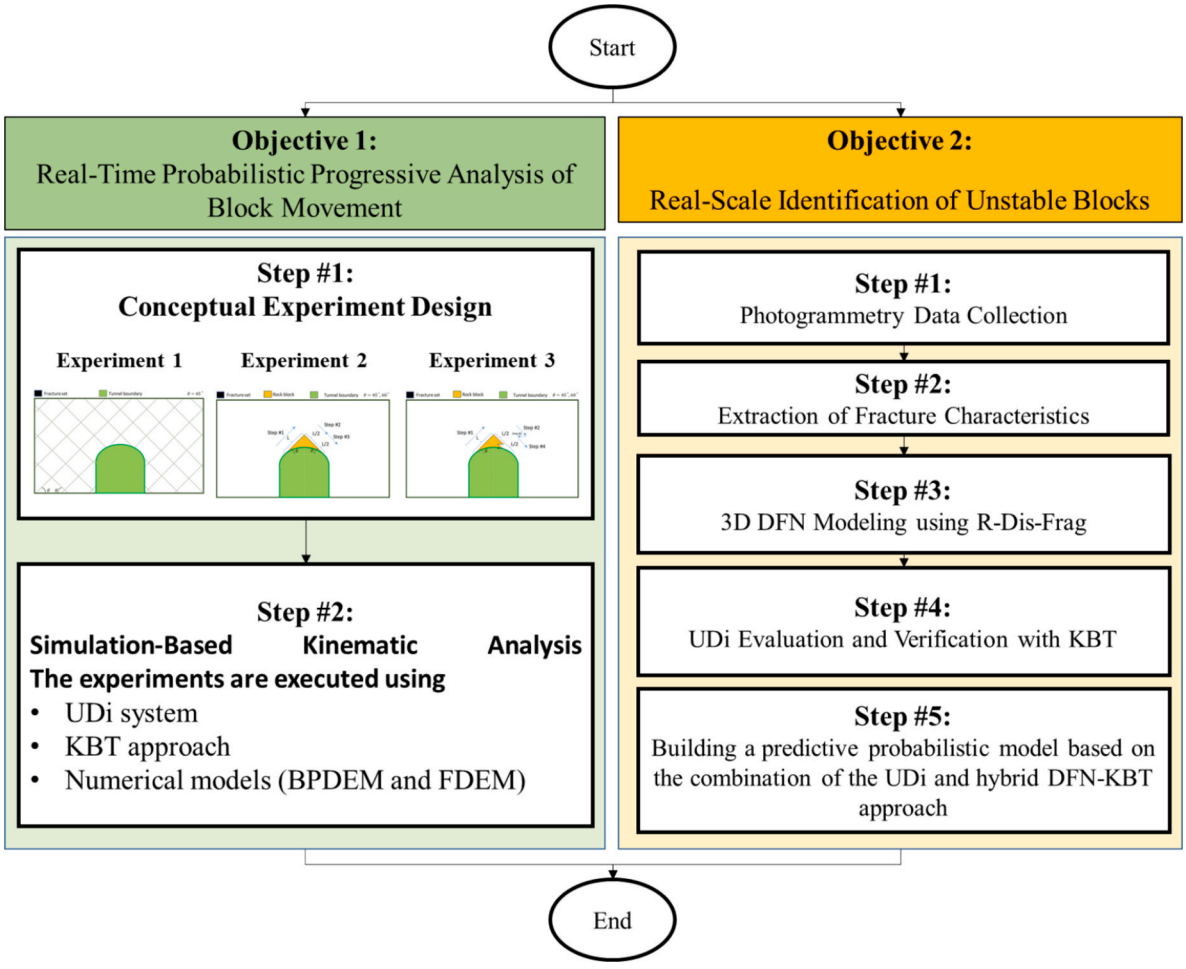


Fig. 2. Flowchart of the study.

the hybrid DFN-Key Block Theory (KBT):

Step #1: Photogrammetry Data Collection: A photogrammetric data bank is compiled from a case study, capturing detailed surface imagery of exposed rock faces in tunnels or slopes.
Step #2: Extraction of Fracture Characteristics: The main discontinuity sets—orientation, spacing, length, persistence—are extracted from the 3D data using specialized tools, forming the structural input for modeling.
Step #3: 3D DFN Modeling using R-Dis-Frag: The R-Dis-Frag code is used to construct a realistic Discrete Fracture Network (DFN) model that represents the spatial distribution and geometric properties of fractures.
Step #4: Automated UDi Evaluation and Verification with probabilistic DFN-KBT approach: The UDi system is applied to identify unstable rock blocks. Its predictions are cross-validated with results from the probabilistic DFN-KBT approach to assess consistency and improve confidence in UDi's hazard zonation capabilities.
Step#5: building a predictive probabilistic model based on the combination of the UDi and hybrid DFN-KBT approach

This objective corresponds to the field-based validation case, emphasizing static block identification and hazard mapping, while demonstrating UDi's capability to construct fast and reliable predictive models of rock block failure.

To provide the necessary context, we first briefly introduce the fundamental concepts of key block theory, the Bonded Particle Discrete Element Method (BPDEM), and Finite-Discrete Element method (FDEM) the experimental dataset used in this study.

4. Theoretical and numerical simulated based validation

4.1. A) Key block theory (Warburton's method)

The Key Block Theory involves first considering a rigid rock block that can only translate and not rotate during movement. Additionally, any surrounding fixed face must remain parallel to its mating rock block face during the possible block movement. As shown in Fig. 3, the block's movement is restricted by the block's geometry and the forces acting on it, which are described by **Equations (2)** and **Equation (3)**:

$$\widehat{n_1}.\widehat{m}0 \tag{2}$$

$$R.\widehat{m} > 0 \tag{3}$$

Where $\widehat{\boldsymbol{m}}$ is a unit vector representing the direction of the block movement, $\widehat{n_1}$ is the unit normal of the fixed face 1 directed out of the face, and $\widehat{R}$ is the block's weight acting vertically downward on its center of mass.

4.1B) Bonded Particle Discrete Element Method.

The Bonded Particle Discrete Element Method (BPDEM) is a powerful numerical technique used to model the mechanical behavior of geomaterials. In this approach, the material is idealized as a collection of discrete particles interconnected by bonds that replicate the strength and deformability of a continuous solid. These interparticle bonds are capable of transmitting both tensile and shear forces, enabling the simulation of fracture initiation, crack propagation, and complex failure patterns. BPDEM is particularly well-suited for analyzing jointed rock masses and capturing progressive failure processes with high spatial and temporal resolution. Its versatility has made it a widely adopted tool in geomechanics, rock engineering, and tunneling applications, where it is used to investigate the stability of underground excavations and evaluate potential failure mechanisms under various loading scenarios (e.g., Zhang and Wong 2013; Chen et al., 2022; Rasmussen & Min, 2023).

4.1.C) Finite Discrete Element Method.

The Finite-Discrete Element Method (FDEM) is an advanced numerical approach that combines the strengths of the Finite Element Method (FEM) and the Discrete Element Method (DEM) to model the mechanical behavior of fractured and discontinuous materials. In FDEM, the material is initially represented as a continuous medium using finite elements, allowing for accurate simulation of stress distribution, deformation, and elastic behavior. As loading progresses and failure criteria are met, cohesive interactions between elements are broken, enabling the transition from continuum behavior to discontinuum mechanics. This hybrid framework allows FDEM to capture both the onset of damage and the subsequent development of fractures, including fragmentation, crack coalescence, and block separation. FDEM is particularly effective in simulating dynamic fracture processes and complex failure mechanisms in rock masses, making it a valuable tool in rock mechanics, slope stability analysis, and underground excavation modeling (Kang et al., 2022b; Yan et al., 2023; Zhang et al., 2025; Deng and Liu, 2025; Wang et al., 2025).

4.2. Theoretical comparison between universal discontinuity index and key block theory

In this section, to validate the performance of UDi in detecting unstable blocks and unstable zones, at first a comparison is made between its kinematic components and the Key Block Theory (KBT) in different scenarios including different conditions of theta between $\widehat{R}$ and $\widehat{m}$ of the rock blocks around the excavations. For this purpose, two block scenarios, as shown in Fig. 4, have been considered. The block movement type 1 (Fig. 4**a**) represents a block that theta between $\widehat{R}$ and $\widehat{m}$ is 0 – easy to identify as unstable– on the crown of a D-shape tunnel in a rock mass with 2 main joint sets. The block movement type 2 (Fig. 4**b**) represents a block that theta between $\widehat{R}$ and $\widehat{m}$ is between 0 and 90 – complex to identify as unstable – on the crown of a D-shape tunnel in a rock mass with 3 main joint sets.

As specified in **Equation (2)**, the condition for block instability is that the rock block must either move away from the fixed face or translate in a parallel direction. Mathematically, this is represented by the angle theta between the vectors "m" and "n" being between [0°, 90°). Similarly, **Equation (3)** determines that the block movement is in a direction where the force has a positive component.

Mathematically, this is represented by the angle theta between the vector's "m" and "R" (which is always in the vertical direction) being between (0°, 90°).

In case a range of directions becomes compatible with both restrictions (0° < theta < 90°), **Equation 4** guarantees a unique $\widehat{m}$ in which the component of the driving force is maximum:

$R.\widehat{m} = max($ Subject to **Equation (2)** and **Equation (3)** (4).

In the UDi system, the factor C.V. tot-ver (total circular variance with respect to the vertical direction) accounts for these conditions, as

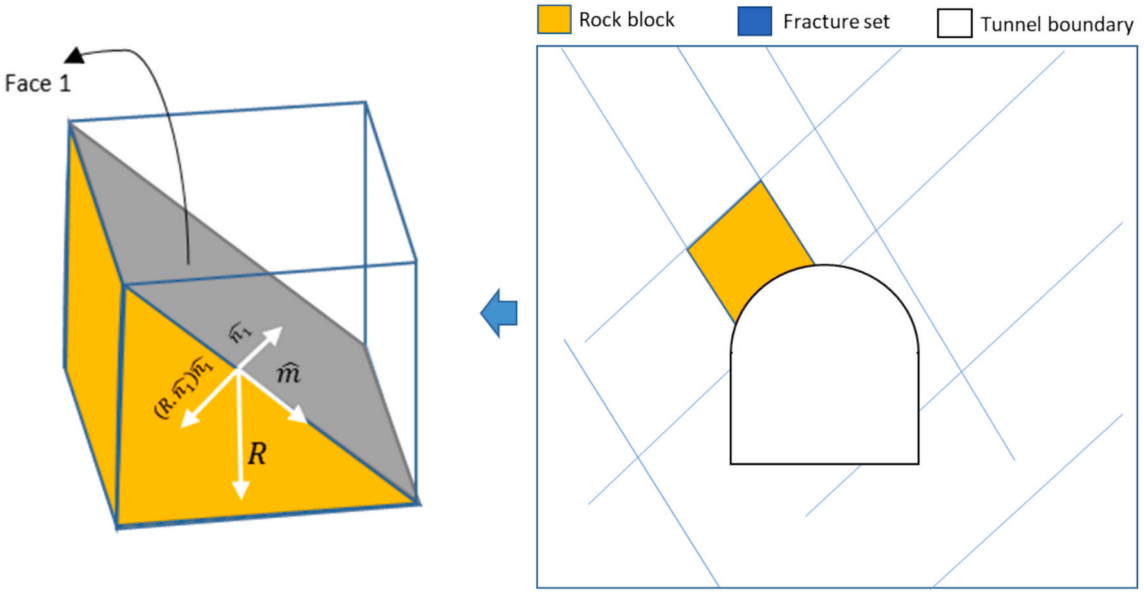


Fig. 3. Components of the KBT approach presented by Warburton (1981).

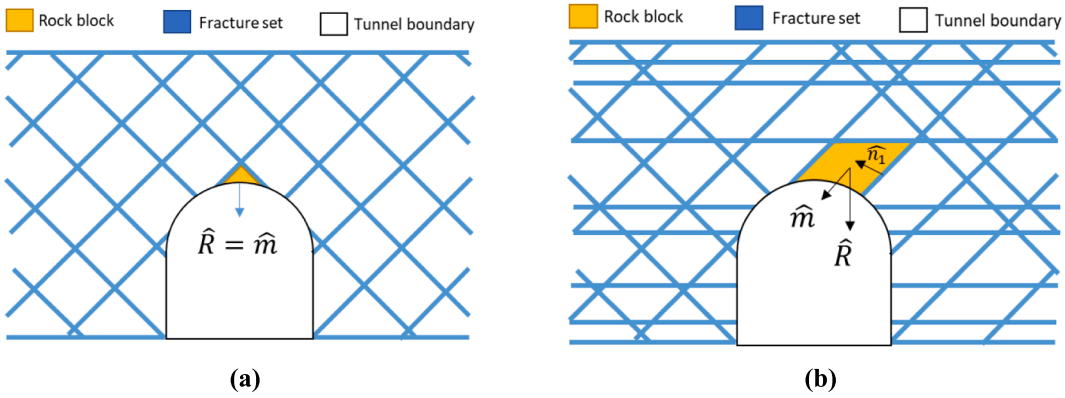


Fig. 4. Schematic presentation of designed scenario for comparison between UDi and KBT approach a) block movement type 1, b) block movement type 2.

expressed in **Equation (5)** below.

$$C.V_{tot-ver} = \frac{1}{2N}\sum_{i=1}^{n}|R + n_i| = \frac{1}{2N}\sum_{i=1}^{N}\sqrt{2 + 2 \times cos\theta_i} \quad (5)$$

(1) If θ_i belongs to [0, 90] then $\cos(\theta_i)$ belongs to [0,1] meaning that C.V. $_{tot\text{-}ver}$ belongs to $[\sqrt{2}/2, 1]$
(2) If θ_i belongs to (0, 90] then $\cos(\theta_i)$ belongs to (0,1] meaning that C.V.$_{tot\text{-}ver}$ belongs to $(\sqrt{2}/2, 1]$
(3) Condition (3) means that C.V.$_{tot\text{-}ver}$ belongs to $(\sqrt{2}/2, 1]$ interval.

The second step, identifying the type of block movement, begins by assuming that the block is initially free, and fixed faces will be added sequentially to the model's geometry. Where there is no fixed face, the block's movement is in the same direction as the driving force (Fig. 4**a**), representing the first movement type in **Equation (6)**;

$$\hat{R} = \hat{m} \quad (6)$$

In this case, the block is free to fall, meaning there is no resisting component acting against the vertical direction of motion. Mathematically, this implies that the force vector **m** is aligned with the resultant vector **R**, i.e., **m** = **R**. If the vertical downward direction is defined as **(0, −1, 0)**, then vector **m** is also **(0, −1, 0)**.

This condition can be expressed mathematically as $\theta i = 0$, resulting in C.V. $_{tot\text{-}ver} = 1$. In other words, when the C.V. $_{tot\text{-}ver}$ equals 1, the block is free to fall.

Adding a fixed face to the condition would not affect the block's movement in case its direction satisfies **Equations 3** and **4** and also **Equation (2)**, which is expressed by considering this new face, i.e. face 1, in **Equation (7)**;

$$\hat{n}_1.\hat{m}0 \quad (7)$$

As in type 1 where $\hat{R}$ is equal to $\hat{m}$, **Equation (7)** can be restated as **Equation (8)** by substituting $\hat{m}$ with $\hat{R}$:

$$\hat{n}_1.\hat{R}0 \quad (8)$$

Conversely, if $\hat{n}_1.\hat{R} < 0$, the fixed face will either prevent block movement or deflect it in a new direction. In this scenario, the C.V. belongs to the interval between zero and $\sqrt{2}/2$.

In the latter case, which is the second type (Fig. 4**b**), the block's movement must still satisfy the restrictions of **Equations (6), 3, and 4**.

The new $\hat{m}$ is in the steepest direction $(\widehat{f1})$ on the fixed face, which is equal to the direction of the orthogonal projection of R on the plane of the fixed face. Therefore, **Equations (9) and (10)** represent the second type of block's movement:

$$\hat{f}_1 = R - (R.\hat{n}_1)\hat{n}_1 \quad (9)$$

$$\hat{m} = \hat{f}_1 \quad (10)$$

In this case, the total C.V. (equivalent to vector m) is equal to the maximum value of the square root of the ith component of sigma $(1 + \cos\theta_i)$, which corresponds to the steepest discontinuity plane.

The two test cases show that UDi reproduces the key-block kinematic checks while adding a continuous risk scale. When the block's weight vector aligns with its release path (Type 1), UDi's circular-variance term reaches its maximum (**C.V._tot-ver = 1**), matching KBT's "immediately unstable" verdict.

- **Type 1 block** (two joint sets, θ (R,m) = 0°). Warburton → "immediately unstable". UDi → CV $_{tot\text{-}ver} = 1$.

Introducing a third joint set (Type 2) forces a deflected release; UDi automatically drops **C.V._tot-ver** into the $\sqrt{2}$ / 2–1 range, signaling partial restraint rather than outright stability.

- **Type 2 block** (three joint sets, 0°<θ < 90°). Warburton → "conditionally unstable". UDi → C.V. $_{tot\text{-}ver} \in (\sqrt{2}/2,1)$ less than 1.

This conditional instability is governed by the geometry of the rock block and the friction angle of the bounding fractures, which provides resistance to movement (Amirkhani et al., 2025). Accordingly, the Universal Discontinuity Index (UDi) not only replicates Warburton's binary (stable/unstable) classification but also extends it into a continuous stability spectrum, enabling integration into probabilistic hazard maps.

To illustrate this in detail, we apply Warburton's vectorial approach step by step. Specifically, we analyze the stability of a square rock block resting on an inclined plane, with one free face and two constrained opposing faces (see Fig. 5).

In the first step, the vectorial sum of the contributing unit normal results in a direction aligned with unit vector $\mathbf{n_2}$ $(\mathbf{n_1} + \mathbf{n_2} + \mathbf{n_3} = \mathbf{n_2})$, which represents the net driving force direction. In addition, no one of the unit vectors have a component against driving force m which mean kinematically is feasible. Since the angle between (driving force) and the weight vector is 60°, the wedge is potentially unstable—its movement is primarily governed by the frictional resistance mobilized along the fracture surfaces.

Moreover, the UDi preserves the failure directionality failure directionality or block release mechanics in complex geometries according to following:

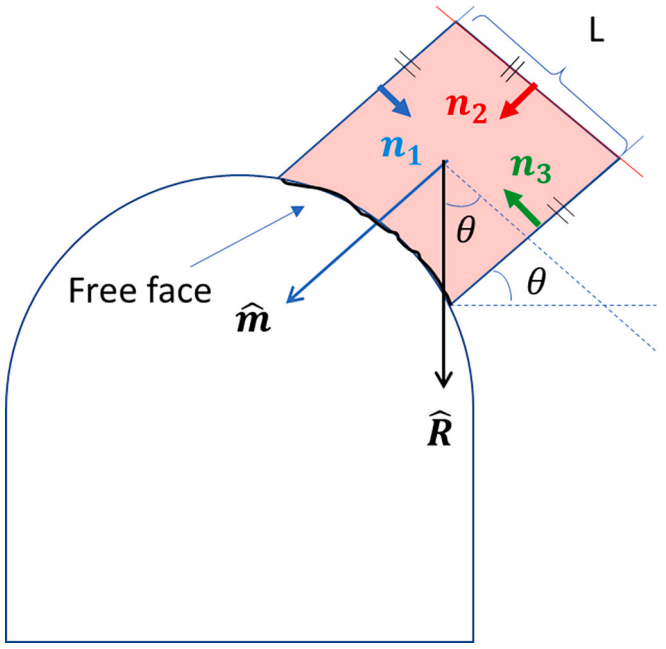


Fig. 5. Geometric configuration of a square rock block resting on an inclined plane with one free face and two constrained opposite faces, representative of a block located between the roof and wall of a tunnel.

- **Vector consistency.** C.V. $_{\text{tot-ver}}$ is computed from the **vector sum R** + $\mathbf{n_i}$ for every plane i; therefore each fracture's contribution is evaluated in its *true spatial orientation*.
- **For Complex geometries** Equations (7)–(10) in the manuscript define the release (slip-or-fall) vector **m** as the orthogonal projection of the driving vector **R** (usually gravity or the resultant of gravity + stress) onto the candidate failure plane with normal **n1.**

$$\hat{f}_1 = R - (R.\hat{n}_1)\hat{n}_1, m = \hat{f}_1$$

- If $\mathbf{f_1}$ is non-zero and points out of the excavation (daylights), the block can slide or topple in that exact direction.
- The **circular-variance term** C.V. $_{\text{tot-ver}}$ then scores how favorable that direction is; the closer m is to R, the larger the score.

Because UDi outputs a scalar from $\sqrt{2}/2 \rightarrow 1$ for every location, directional information can be re-extracted (e.g., mapping the steepest per cell) while still benefiting from the probabilistic scale. The approach therefore **retains the mechanics of Warburton's method** (sliding versus free-fall, deflection by fixed faces) and extends the analysis to jointed media with many interacting planes, without losing directional resolution.

4.3. Probabilistic batch progressive block movement analysis

Structurally controlled failure is a complex process influenced by spatial variability in geological features and dynamic stress redistribution during excavation. Rock mass conditions—including discontinuity patterns, faults, and intact rock properties—can vary drastically over short distances, and the resulting stress field evolves as excavation progresses. These factors introduce significant uncertainty into stability assessments.

Traditional models, such as Warburton's Key Block Theory (KBT), are primarily designed to identify kinematically removable blocks that are directly adjacent to the excavation boundary. While effective for detecting immediate, single-event instabilities, classical KBT does not account for the progressive or batch failure of rock blocks—where the initial detachment of key blocks can trigger the subsequent destabilization and failure of surrounding blocks. Capturing such sequential failure mechanisms requires iterative extensions of KBT, as developed in more recent studies (Zhu et al., 2024), which go beyond its original deterministic formulation.

The Universal Discontinuity Index (UDi) addresses this limitation by providing a probabilistic framework that captures both isolated and sequential block instabilities. This capability has been validated through theoretical development and numerical modeling. As illustrated in Fig. 6, UDi's performance was tested against KBT, Finite Discrete Element Method (FDEM), and Bonded Particle Discrete Element Method (BPDEM) in a tunnel scenario involving two intersecting joint sets. The comparative results demonstrate UDi's superior ability to represent the evolving behavior of rock blocks under progressive failure.

Fig. 7 illustrates a conceptual comparison between KBT and UDi. According to KBT, only blocks labeled "1″ are identified as unstable due to their direct exposure. However, this narrow view ignores blocks "2" and "3," which may become unstable as a consequence of the failure of block "1," highlighting the cascading nature of progressive failure. KBT, limited to immediate contact geometry, cannot anticipate such deeper instabilities.

UDi overcomes this by incorporating a kinematic factor based on the interaction between fracture density (P32) and the circular variance relative to the vertical direction (CV.tot-ver). This parameter reflects the spatial orientation of fractures with respect to the principal driving force, enabling a comprehensive assessment of instability risk. In the example shown, with fractures oriented at 45° and 12 traces within the zone of interest, the calculated CV.tot-ver is 0.92—indicating a high probability of kinematic failure, especially when combined with a high P32 value.

Unlike KBT's deterministic, stepwise approach—where each discontinuity is assessed in sequence—UDi treats the fracture network as a unified system. This holistic perspective captures emergent behaviors and interactions that are essential for understanding failure in complex rock masses. Therefore, UDi provides a more realistic, scalable, and predictive tool for assessing structural instability, particularly in digital twin applications for underground excavations.

To test the UDi-based forecasts generated for Experiment 1, we performed two independent numerical studies: a Bonded-Particle Discrete Element Model (BPDEM) and a Finite-Discrete Element Model (FDEM). In this study, the modeling assumptions were based on the conceptual premise that rock blocks behave as perfectly rigid bodies and that joint surfaces are frictionless and cohesionless. Figs. 8 and 9 summarize the simulation workflow and key results, including the geometric configurations, boundary conditions, and resulting block release patterns. While both simulations share identical tunnel dimensions, joint spacing and orientation, and overall model length, the FDEM domain was given a slightly larger vertical extent to reduce the sensitivity of the model to boundary constraints—a known issue in continuum-based solvers. This adjustment had no effect on the observed failure mechanism and allowed both models to reproduce a consistent pattern of progressive block detachment surrounding the tunnel.

BPDEM (Fig. 8). The domain (20 m × 10 m) was filled with rigid spherical particles whose radii follow the distribution shown in Fig. 8**a**. In the BPDEM simulations, block rigidity was enforced by assigning extremely high values (1.0e12) to the parallel bond micro-parameters, including tensile strength (Pb_ten), cohesion (Pb_coh), normal stiffness (Pb_Kn), and shear stiffness (Pb_Ks), and the elastic modulus of the particles (Pb_deform emod). A D-shaped cavity (4 m span, 2 m crown radius; Fig. 8**b**) was excavated, after which two orthogonal joint sets inclined 45° to the horizontal were introduced with the smooth-joint contact model (zero cohesion, zero normal and shear stiffness) to reproduce frictionless sliding (Fig. 8**c–8d**). Joint surfaces were modeled using the smooth-joint contact model, which allows for slip along predefined planes independent of the particle contact geometry. To simulate frictionless, unbonded joints, all smooth-joint micro-parameters—Sj_ten, Sj_coh, Sj_Kn, and Sj_Ks—were set to zero. With

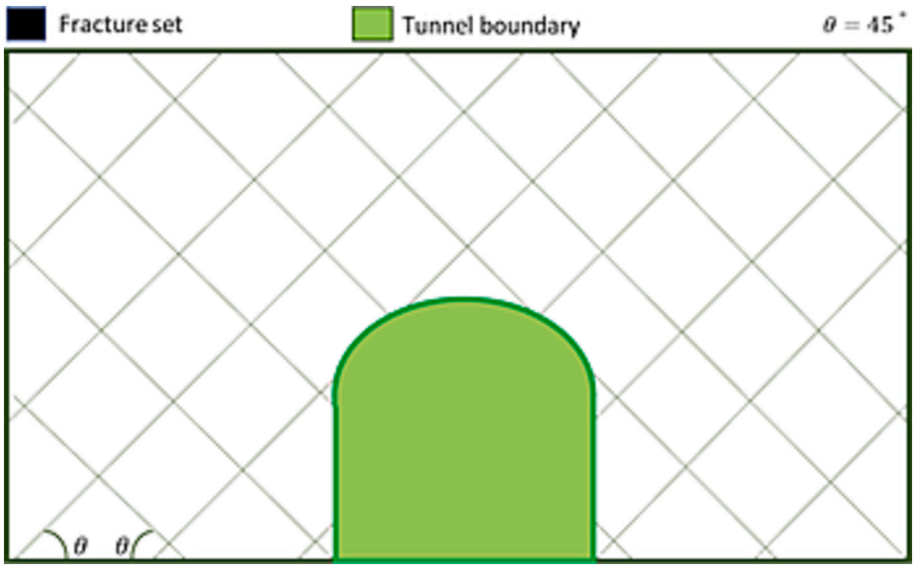


Fig. 6. Schematic of Experiment 1: Tunnel scenario in a rock mass with two joint sets.

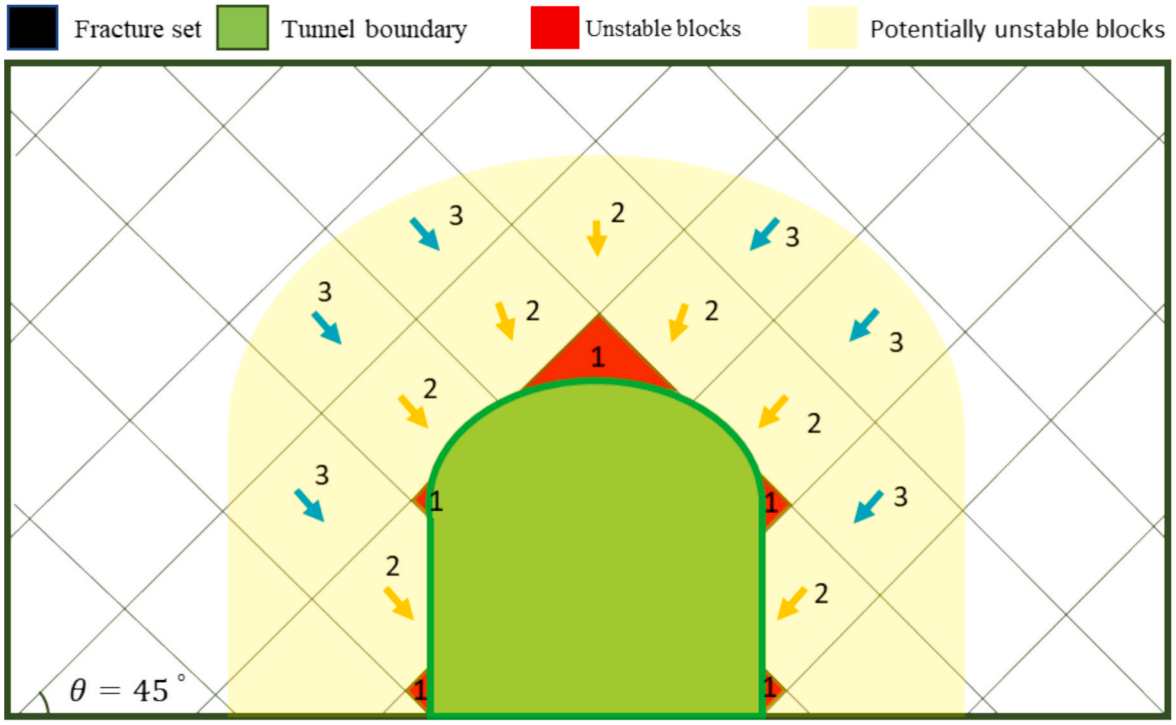


Fig. 7. Illustration of the limitation of KBT in capturing batch progressive failure. Blocks labeled "1″ are detected by KBT; their failure may trigger instability in blocks "2" and "3.".

gravity as the sole driving load, the simulation captured three successive waves of block detachment and descent (Fig. 8**e**).

*FDEM (*Fig. 9*)*. An equivalent rigid continuum was discretized into uniform tetrahedral zones (Fig. 9**a**) and intersected by the same joint sets (Fig. 9**b**). In the FDEM simulations, rigid blocks were represented as linear elastic continua with a Young's modulus of 1e12 Pa, cohesion and tensile strength 1e8 Pa. Fracture interfaces were defined using broken (closed and cohesion less) models with zero friction to match BPDEM assumptions. These values ensured that the blocks remained intact under stress while permitting separation along joints as required. In both models, the rock mass density was uniformly assigned as 2700 kg/m3. Progressive displacements reveal two distinct release stages followed by continuous run-out of unstable blocks (Fig. 9**c–9d**). The following sections compare the results of each numerical model with the UDi-based predictions (see Fig. 7) in terms of the progressive formation of mobile blocks.

Both numerical schemes reproduced the sequence, location, and timing of block failures predicted by the UDi's kinematic component, confirming that the index correctly identifies zones prone to batch-progressive instability even under a gravity-only, frictionless idealization.

To clarify the comparison between UDi and numerical results, emphasis was placed on matching spatial trends and failure sequences rather than achieving numerical equivalence. Given the differing

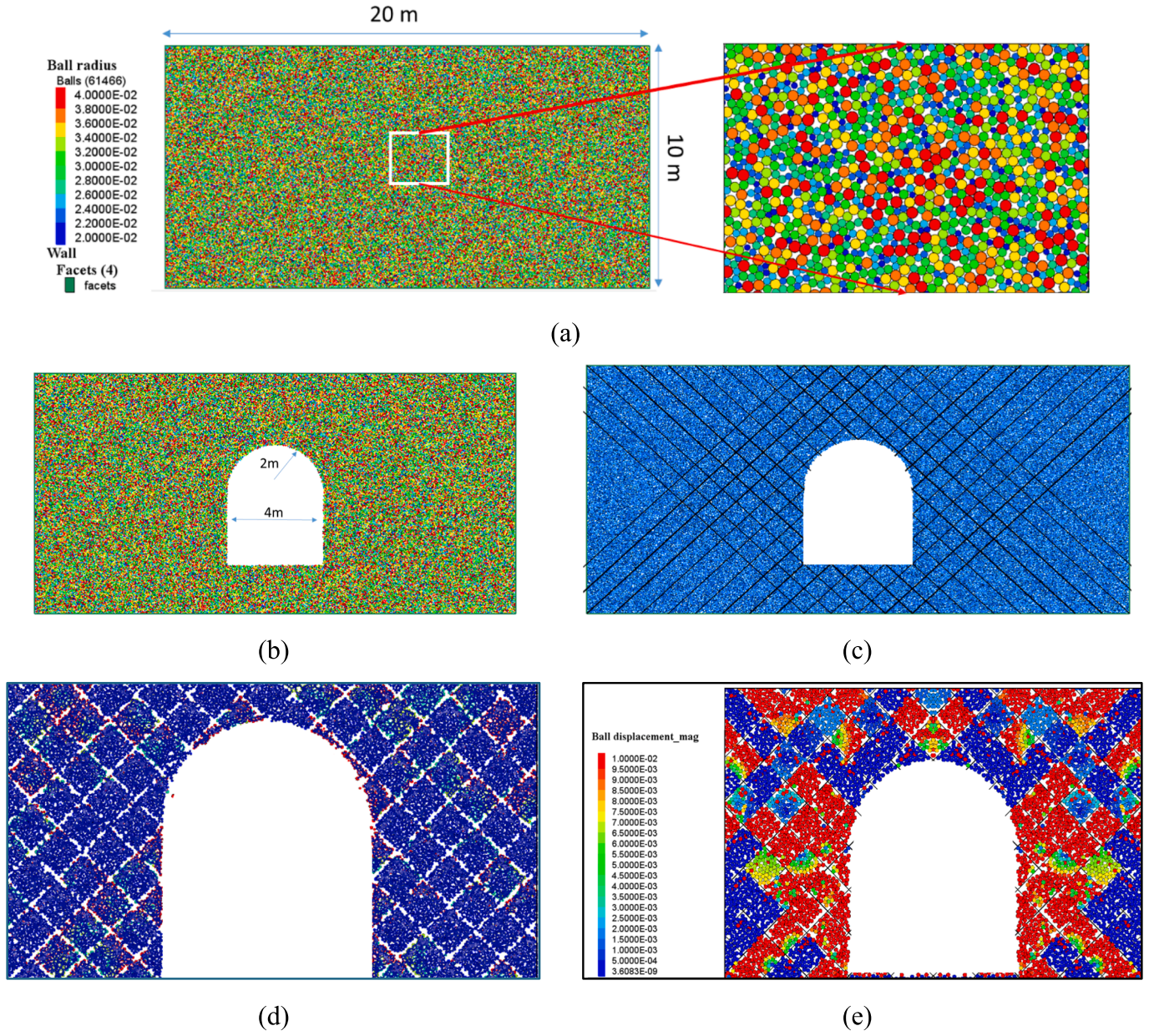


Fig. 8. Step-by-step Simulating the Experiment 1 using BPDEM; a) size distribution of rock material in the initial equilibrium, b) excavating the D-shape tunnel, c) applying joint sets d) formed blocks near the tunnel; e) movement of the first, the second and the third layer of unstable blocks.

formulations and resolutions of BPDEM and FDEM, their outputs were analyzed separately and qualitatively compared with the UDi hazard map. UDi scores were independently normalized to allow fair visual interpretation. Despite these methodological differences, consistent zones of progressive block failure were identified across all approaches, reinforcing the theoretical validity of the kinematic UDi framework.

The close match between UDi scores and the BPDEM/FDEM outcomes demonstrates three practical advantages:

1) Accuracy under minimal input. Even with no applied in-situ stresses and friction-free contacts, UDi pinpointed the same failure fronts captured by two high-fidelity solvers, highlighting its robustness when data are sparse or uncertain.
2) Computational efficiency. Whereas BPDEM and FDEM required full dynamic simulations, UDi delivered equivalent hazard maps in milliseconds and can be embedded directly in digital-twin pipelines for real-time decision support.
3) Scalability. Because UDi is calculated from routinely collected fracture data, it can be deployed across extensive tunnel networks without the prohibitive cost of large-scale DEM meshes.

These findings confirm UDi as a fast, reliable, and scalable metric for predicting batch-progressive rock-block failure in complex jointed media.

4.4. Fracture propagation-informed refinement of the universal discontinuity index (UDi)

This section demonstrates the ability of the Universal Discontinuity Index (UDi) to capture the stepwise progression of fracture development and its impact on block kinematic instability. The UDi results are compared against Key Block Theory (KBT) and two numerical methods—Bonded Particle Discrete Element Method (BPDEM) and Finite Discrete Element Method (FDEM)—to evaluate consistency and

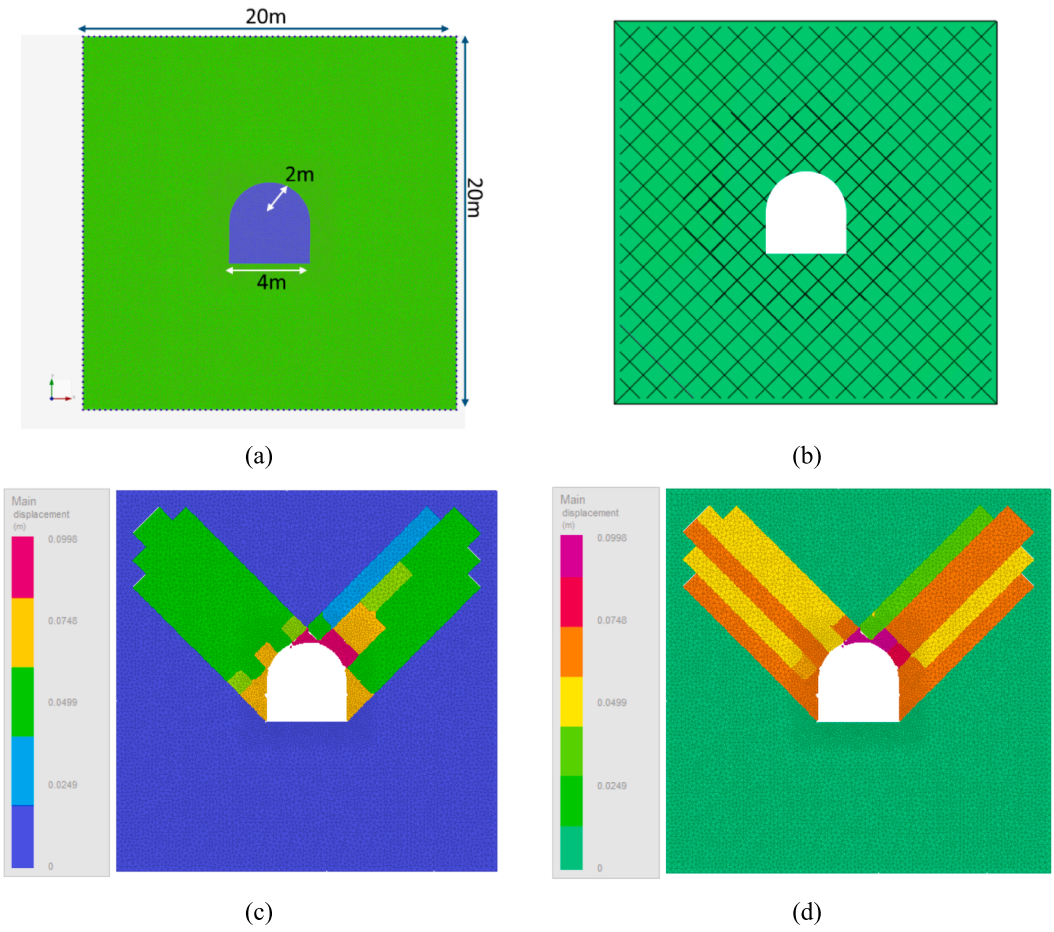


Fig. 9. Step-by-step Simulating the Experiment 1 using FDEM; a) boundary conditions and mesh size distribution around tunnel, b) excavating the D-shape tunnel and shaped blocks result in 2 joint set, c) movement of the first, and the second layer of unstable blocks d) continuously movement of unstable blocks.

predictive performance.

4.4.A) Experiment Design: Wedge-Shaped Block (Fig. 10).

A simplified wedge-shaped block is modeled at the tunnel crown, formed progressively in three stages using two fracture inclinations ($\theta = 45^{\circ}$ and $\theta = 60^{\circ}$). Each stage reflects a new fracture step (Step 1 to Step 3), eventually leading to a fully formed block. The UDi is applied at each step to quantify the evolving kinematic instability.

Numerical Setup (Figs. 11 and 12).

Two types of simulations were conducted:

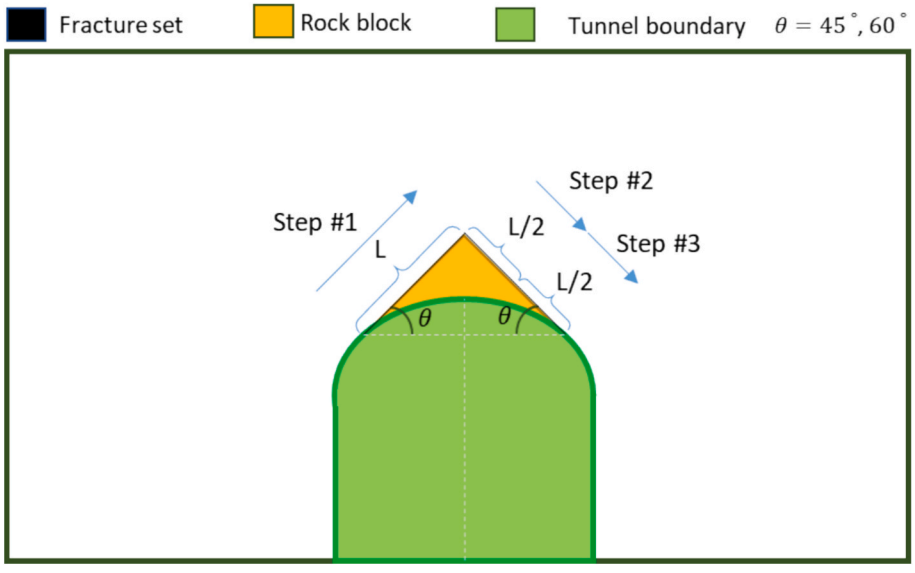


Fig. 10. Schematic illustration of experiment 1, a 2D tunnel sequence with simple wedge-shape rock block.

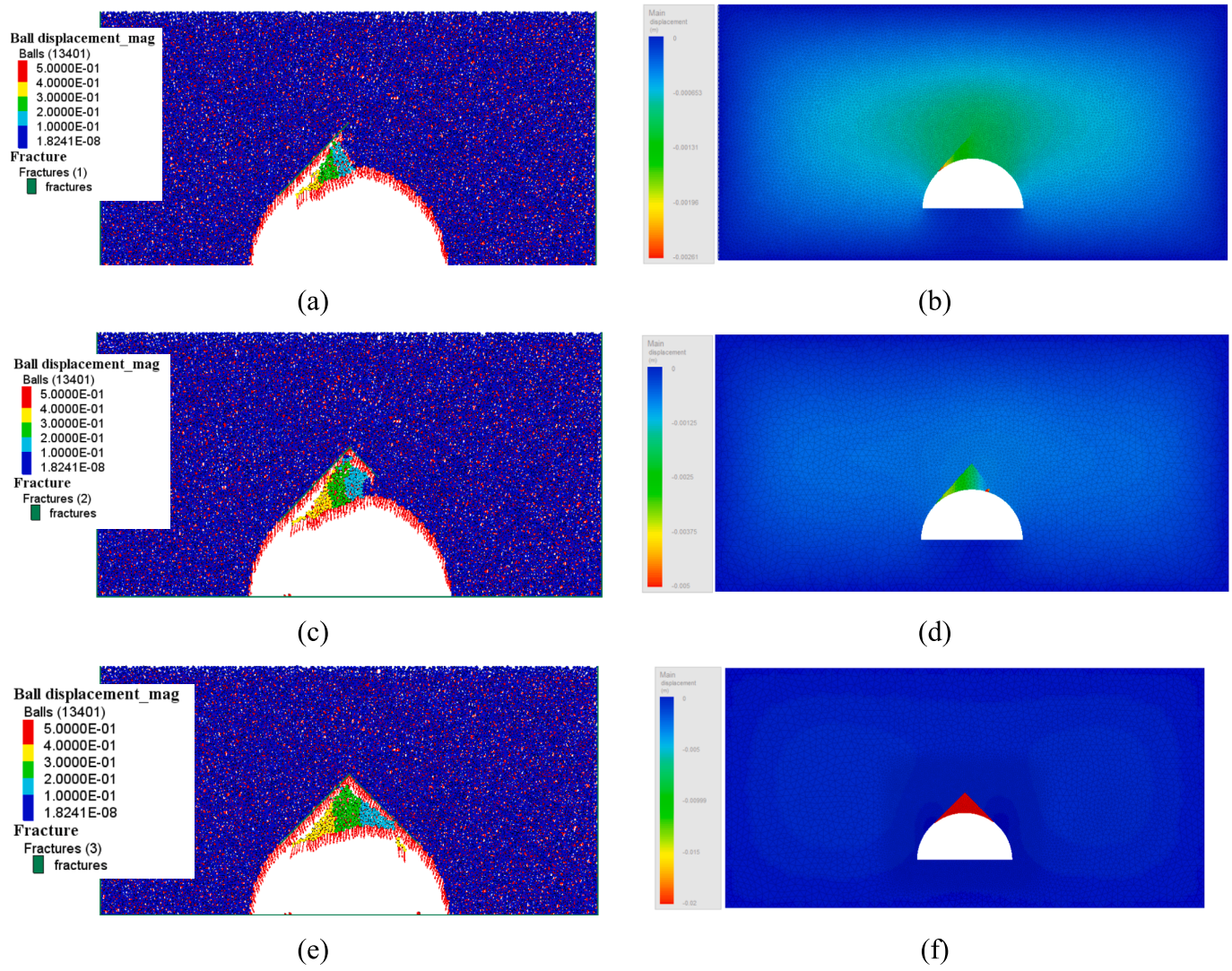


Fig. 11. Step-by-step Simulating the Experiment 2 (45 degree) using DEM (a,c,d) and FDEM (b,d,f); a,b) fracture step 1, c,d) fracture step 2, e,f) fracture step 3 or closed block.

- **BPDEM** (Fig. 11a,c,e; Fig. 12a,c,e) using spherical particles.
- **FDEM** (Fig. 11b,d,f; Fig. 12b,d,f) with an equivalent 2D continuum mesh.

The tunnel cross-section is 4 m in diameter. The surrounding rock is modeled after calcite with UCS = 32 MPa, E = 2.2 GPa, and $\nu = 0.3$. No external stress field was applied to isolate kinematic effects. Fracture surfaces were defined with zero cohesion and zero stiffness, simulating idealized frictionless conditions. Displacement of the formed block was recorded at each step.

The comparative evaluation of the Universal Discontinuity Index (UDi), Key Block Theory (KBT), and numerical models (BPDEM and FDEM), as summarized in Table 2, reveals key insights into the prediction of progressive rock block instability.

First, UDi effectively captures the stepwise evolution of instability associated with the progressive formation of rock blocks. Unlike KBT—which only recognizes instability once a block is fully enclosed—UDi provides continuous insight throughout the fracture development stages, offering early warnings during the block formation process. This stepwise resolution is critical for real-time hazard assessment in tunneling environments.

Second, the results demonstrate a strong correlation between UDi values and average block displacements obtained from the numerical models. Both BPDEM and FDEM simulations show incremental increases in displacement across fracture steps, which align closely with UDi's kinematic scores. This consistency confirms UDi's reliability as a geometry-driven, probabilistic predictor of block instability.

Third, UDi exhibits sensitivity to fracture geometry. While KBT treats blocks formed by 45° and 60° joint sets as equally unstable once fully formed, UDi—and both numerical methods—correctly identify the wedge formed at 60° as more unstable than the one at 45°, as clearly shown in Figs. 13 and 14. This distinction is vital for prioritizing support in structurally complex zones.

4.4.B) Complex Wedge at the Tunnel Crown.

A four-stage wedge is generated at the tunnel crown by fractures dipping 45° or 60° (Fig. 15). Warburton's Key-Block Theory (KBT) classifies the block as stable because no simple release path exists (Fig. 16, red face). However, when the block's self-weight is considered, the Universal Discontinuity Index (UDi) assigns a high kinematic-instability score (0.87), reflecting a realistic failure risk that KBT overlooks.

To validate UDi predictions, the same rock and discontinuity properties from previous tests were used. The simulations were performed using BPDEM and FDEM, focusing on 45° and 60° fracture inclinations. The stepwise evolution of the wedge and its displacement under gravity loading are illustrated in Fig. 17 (45°) and Fig. 18 (69°).

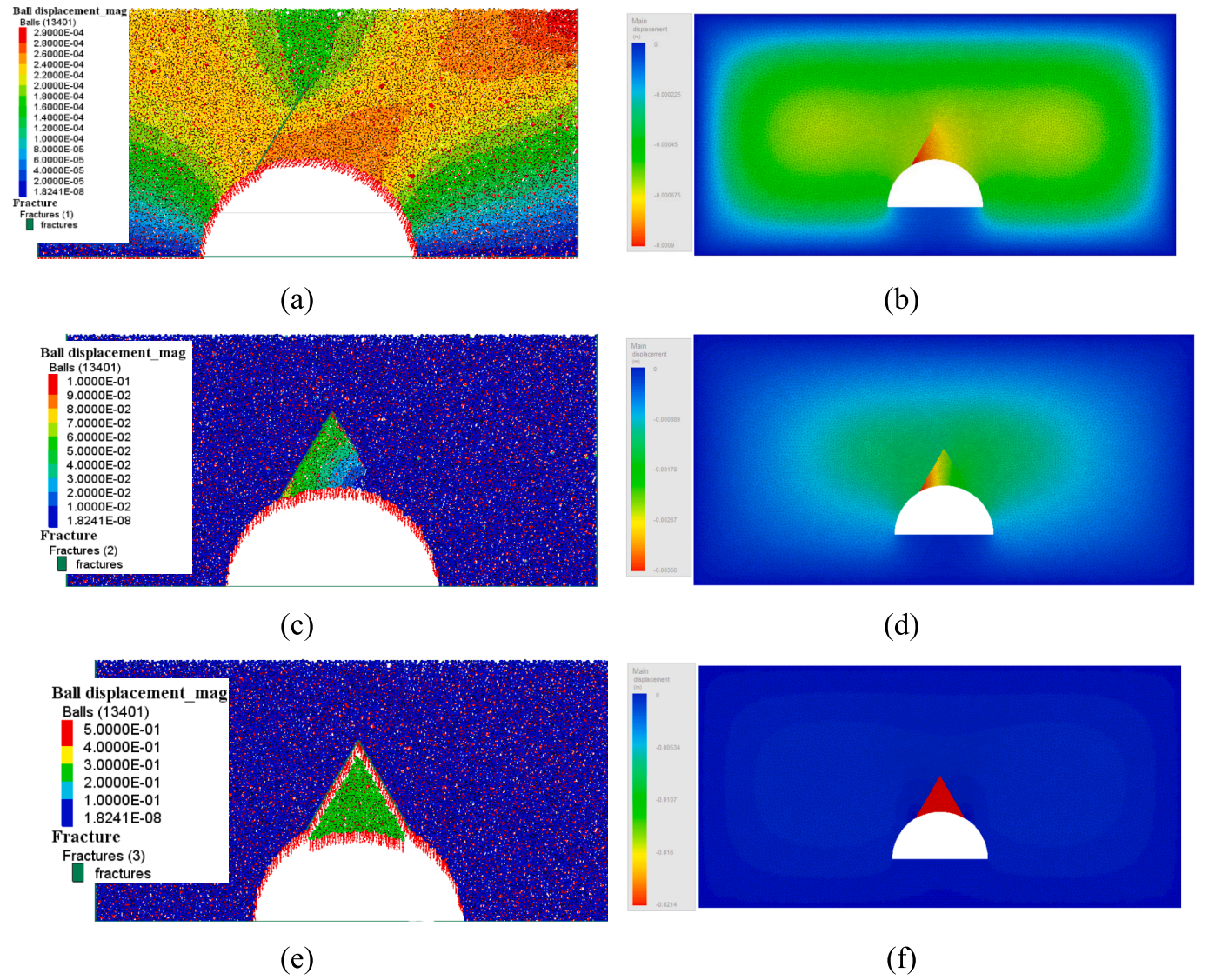


Fig. 12. Step-by-step Simulating Experiment 2 (60 degree) using DEM (a,c,d) and FDEM (b,d,f); a,b) fracture step 1, c,d) fracture step 2, e,f) fracture step 3 or closed block.

Table 2
The results obtained from UDi, KBT, BPDEM and FDEM in probabilistic progressive rock block failure- simulation based Experiment 2.

Theta	Steps of simulation	KBT Decision	UDi Kinematic Value	DEM Average displacement formed block (m) in the same steps number	FDEM Average displacement formed block (m) in the same steps number
45	1	No block created	5.84	0.1	0.00131
	2	No block created	9.97	0.15	0.003
	3	Unstable block created	14.1	0.25	0.0181
60	1	No block created	5.47	0.0002	0.00085
	2	No block created	9.6	0.06	0.0027
	3	Unstable block created	13.74	0.3	0.02

- For the **45° wedge**, both DEM and FDEM models confirm that instability emerges after the block is fully enclosed. This behavior aligns with UDi predictions and reflects its ability to track failure development during each fracture stage.
- For the **60° wedge**, although instability also occurs post-closure, the displacement rate is lower compared to the 45° case. This geometric sensitivity is captured by UDi but not by KBT, which treats both configurations as equally stable until final closure.

Table 3 and Figs. 19 and 20 present the comparison across all methods. Instability in the numerical models is quantified based on the average displacement of the formed block at each fracture stage. The data reveal three important findings:

- **Progressive Accuracy**: UDi reliably tracks the increase in instability through each fracture stage, while KBT identifies failure only after full block formation.
- **Geometric Sensitivity**: UDi captures the lower mobility of the 60° wedge compared to the 45° one—validated by both numerical methods—whereas KBT fails to distinguish between them.

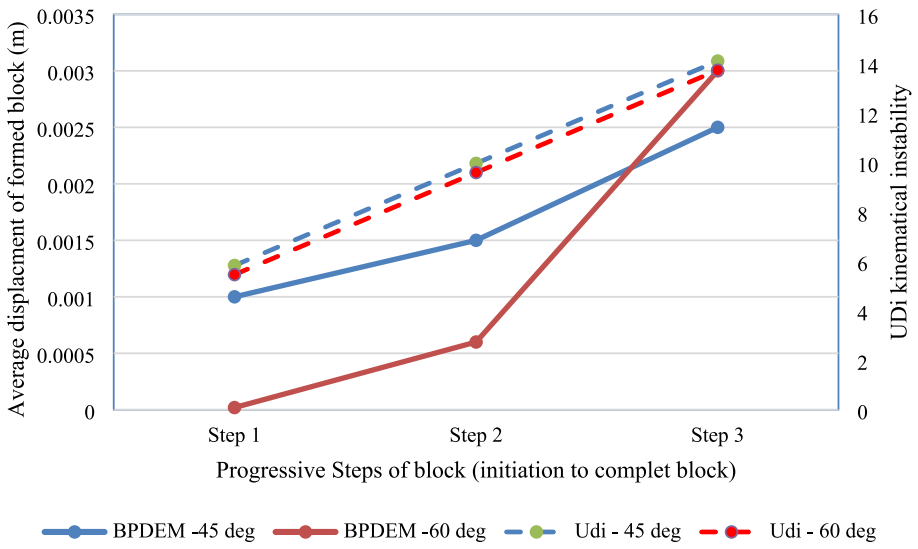


Fig. 13. Application of udi in probabilistic progressive in comparison with the bpdem numerical model- simulation-based experiment 2.

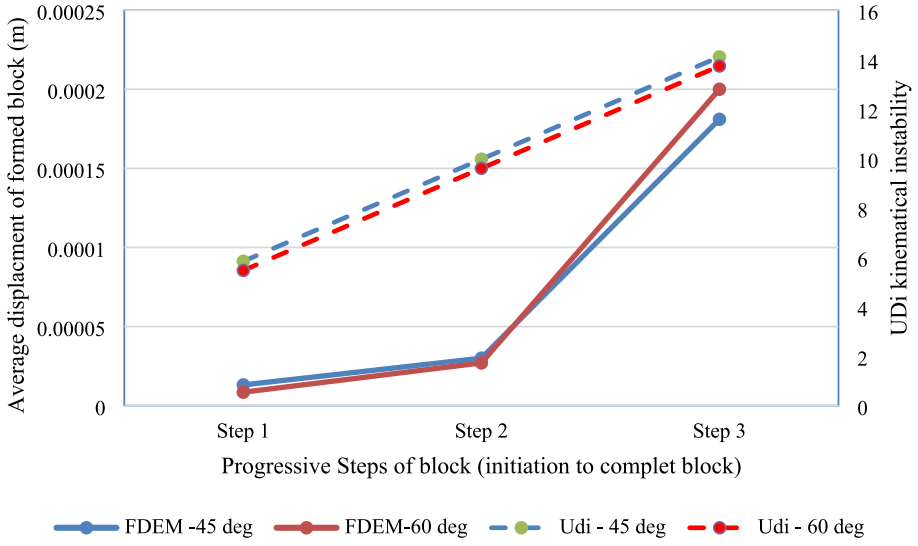


Fig. 14. Application of udi in probabilistic progressive in comparison with the fdem numerical model- simulation-based experiment 2.

- **Strong Agreement**: The displacement trends from BPDEM and FDEM closely mirror UDi's kinematic scores across all stages, confirming its predictive capability.

This experiment confirms UDi's ability to quantify kinematic instability in complex wedge geometries. Unlike classical deterministic models, UDi offers a dynamic and probabilistic perspective that aligns closely with high-fidelity numerical simulations. Its rapid computation and minimal input requirements make it an ideal tool for digital twin frameworks and real-time tunnel hazard assessment.

It is worthwhile to mention that the Universal Discontinuity Index (UDi) exhibits smoother and more consistent trends across different fracture inclinations (e.g., 45° vs. 60°) compared to the BPDEM results, primarily due to fundamental differences in their underlying theoretical frameworks. Similar to FDEM, the UDi is grounded in fracture mechanics principles—particularly the use of stress intensity factors—-where directional dependence on fracture inclination is introduced through bounded trigonometric functions (sine and cosine). As such, when other factors such as material properties and boundary conditions remain constant, changes in inclination alone (e.g., from 45° to 60°) yield relatively moderate differences in computed severity.

In contrast, the BPDEM approach simulates rock as a bonded particle assembly, where fracture initiation and propagation emerge from discrete particle interactions and bond breakage. This micro-mechanical modeling introduces higher sensitivity to fracture orientation, particularly under unconfined or low-confinement conditions, where small geometric shifts can lead to pronounced variations in contact forces, stress redistribution, and damage localization. As a result, BPDEM simulations tend to show larger discrepancies in displacement and failure patterns between 45° and 60° scenarios, as evidenced in Tables 2 and 3.

Nevertheless, the principal aim of this comparison is not to match absolute displacement magnitudes, but rather to assess whether the trend of deformation aligns with progressive damage evolution. In this regard, the UDi consistently reflects the stepwise increase in instability, validating its capacity to capture the evolving nature of damage and its effect on block mobility within the rock mass.

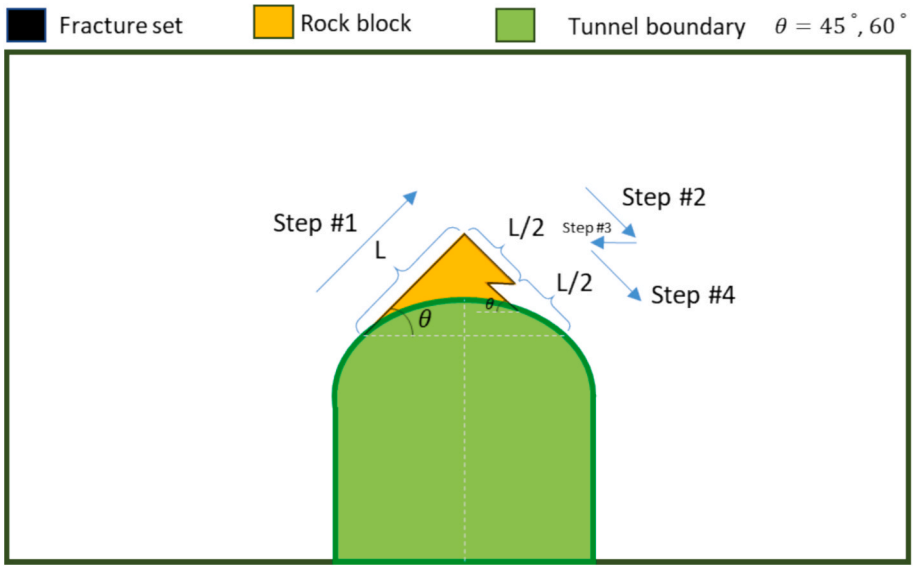


Fig. 15. Schematic illustration of experiment 2, a 2D tunnel sequence with special wedge-shape rock block.

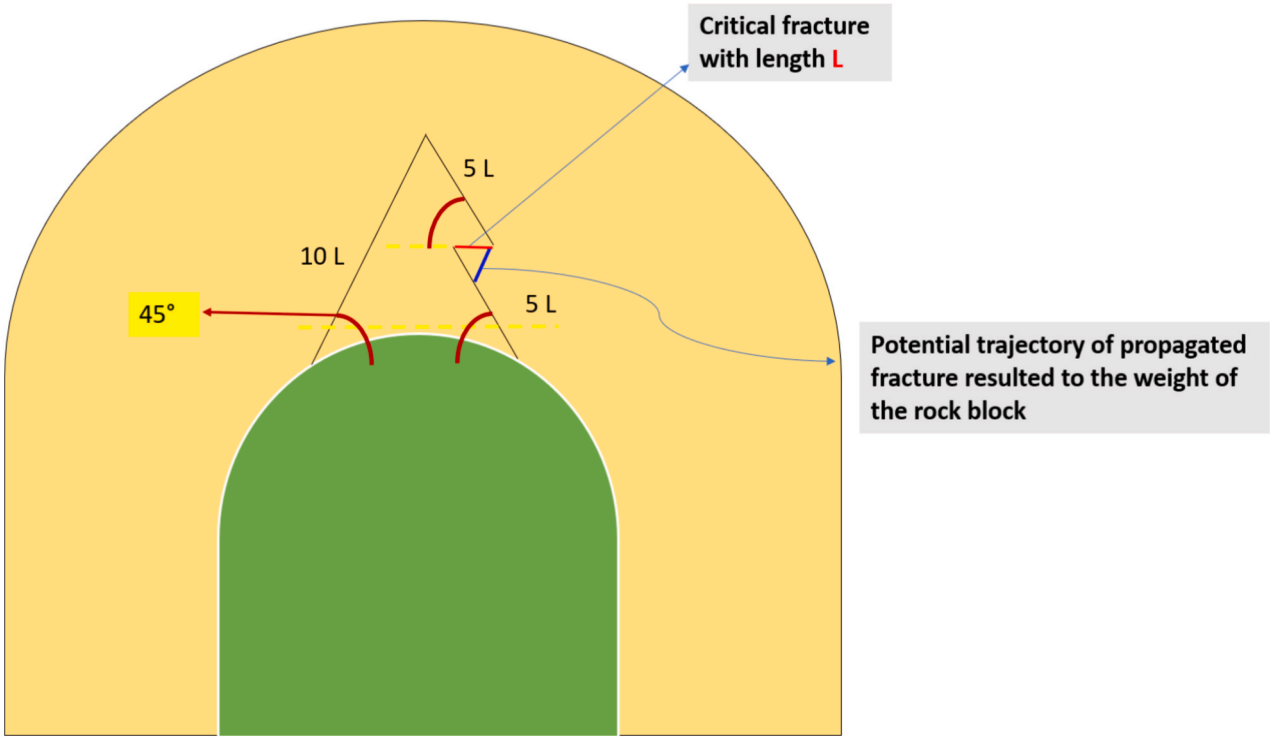


Fig. 16. The conceptual example demonstrates the deterministic sequential impact that Warburton's theory has on the analysis of rock block stability. Red fracture caused the kinematic stability, the blue line shows the potential formed rupture through the weight of the rock block. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5. Delineating the hazardous zones of rock block failure along a El teniente tunnel using UDi

This section aims to apply UDi system for fast predictive hazard maps based on the photogrammetry data set that can be used to identify areas of high risk for rock block failure, thereby allowing for effective risk mitigation measures to be put in place. These maps will provide critical information for decision-making processes related to the planning, design, and construction of tunneling projects in areas where rock block failure is a significant hazard. Also, UDi will be validated through comparing the result with the result of the KBT approach. For this end, an experimental procedure was designed. Fig. 21 illustrates the methodology used to identify high-risk zones for rock block failure, employing the UDi-based approach and Warburton's model. The main steps of this framework encompassed the following steps:

- **Photogrammetry image data gathering**: This involves capturing a series of overlapping photographs of a rock surface, which are later processed to create accurate 3D models or orthophotos used for geological analysis.
- **Extracting fracture sets key characteristics**: This refers to identifying and measuring essential features of rock fractures—such as

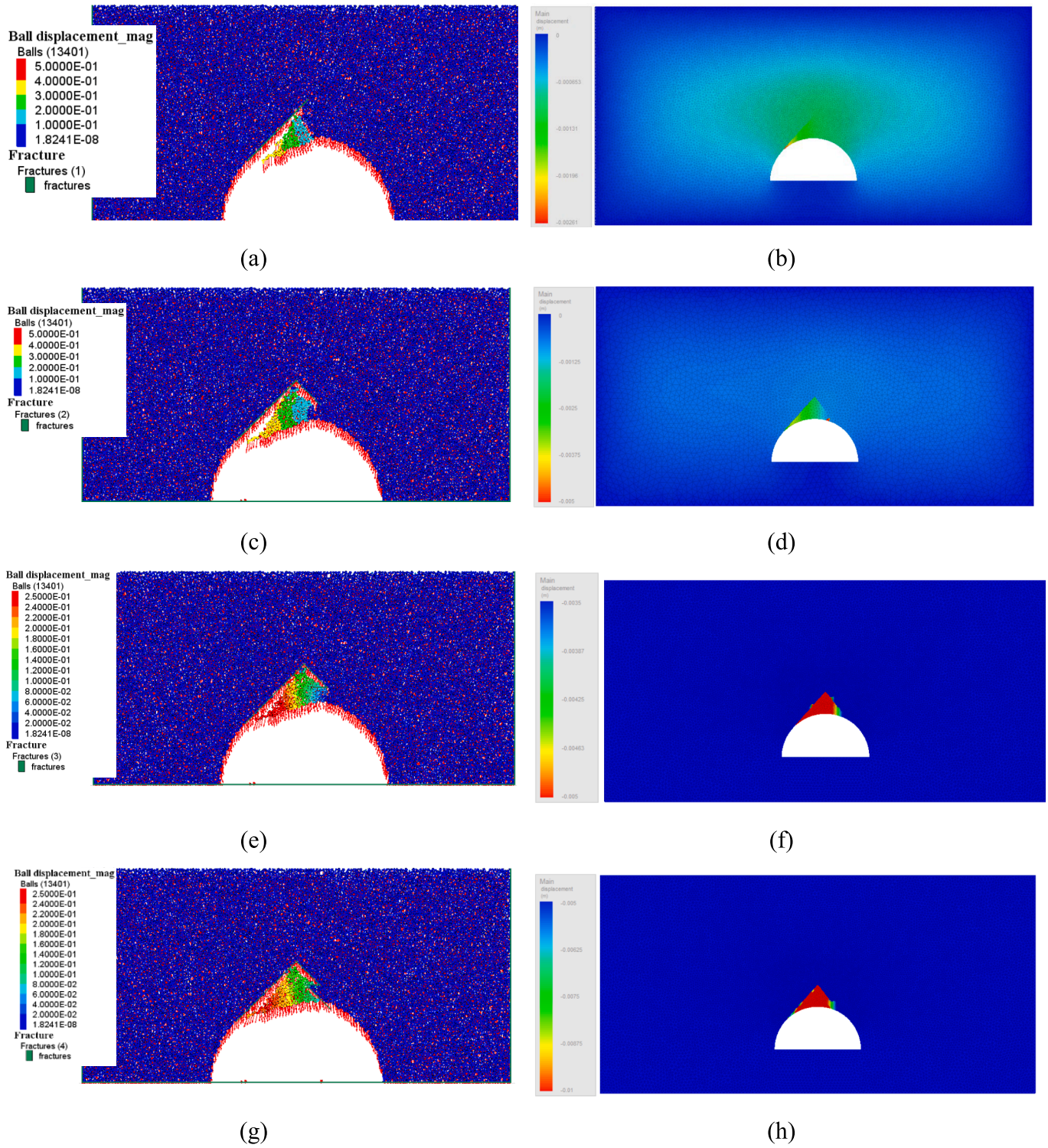


Fig. 17. Step-by-step Simulating the Experiment 3 (45 degree) using BPDEM; a) fracture step 1, b) fracture step 2, c) fracture step 3, d) fracture step 4 or closed specific-shape block.

orientation, spacing, length, and persistence—from 3D models or field data to better understand the structural behavior of the rock mass. In this study the R-Dis-Frag computer package is used to infer discontinuity network parameters (Hekmatnejad and Crespin, 2022).

- **3D Discontinuity System Modeling:** Key parameters for constructing a Discrete Fracture Network (DFN) include the orientation, size, and intensity of discontinuities, which are crucial for developing a stochastic model that realistically represents fracture behavior. Using the R-Dis-Frag software package, a 3D discontinuity network was generated within a cubic domain of 20 m × 20 m × 40 m.
- **Calculating the UDi components:** The components of the Universal Discontinuity Index (UDi)—including volumetric fracture intensity (P32) and total circular variance relative to the vertical direction—are derived from photogrammetry-based data for each tunnel section. To address the inherent uncertainties associated with these

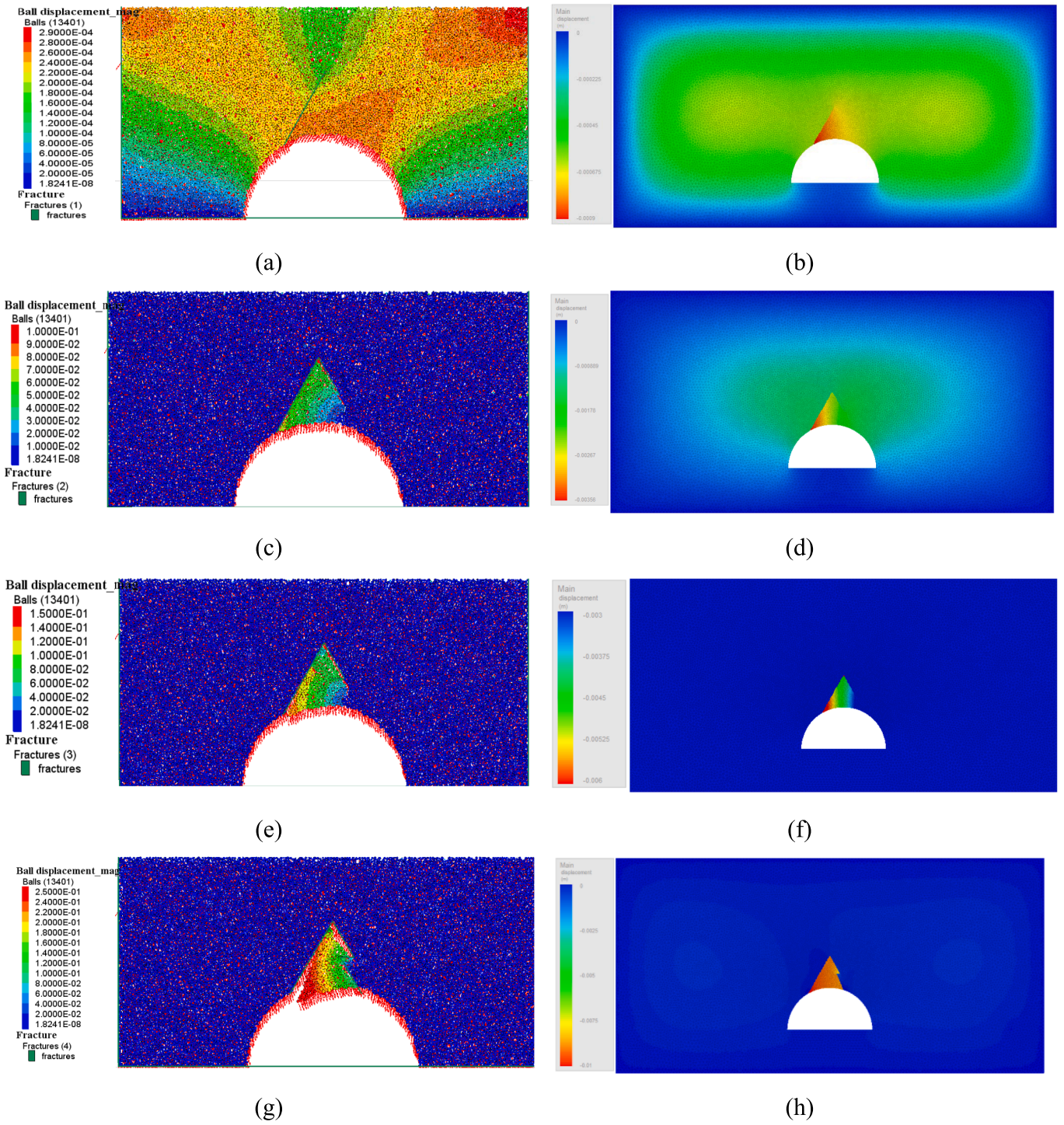


Fig. 18. Step-by-step Simulating the Experiment 3 (60 degree) using BPDEM; a) fracture step 1, b) fracture step 2, c) fracture step 3, d) fracture step 4 or closed specific-shape block.

parameters, the virtual sampling technique proposed by Hekmatnejad et al. (2022) is applied, allowing for a more robust and probabilistic representation of the discontinuity characteristics within the UDi framework.

- **Closed Rock Blocks Geometry Identification:** Determining the geometry of closed rock blocks around the tunnel using R-Dis-Frag code (Hekmatnejad and Crespin, 2022)
- **Application of Warburton's Vectorial-Based Algorithm:** In this step, each rock block is classified as either kinematically stable or unstable. The process involves calculating all active and passive forces acting on the blocks, including the weight vector of the wedges and the resistance force vectors along the block faces, to determine the resultant force vector ($\vec{R}$). It is important to note that the analysis deliberately excludes certain factors, such as water pressure, seismic forces, bolt support, and shear resistance from shotcrete, focusing solely on the primary mechanical interactions influencing block stability.

Fig. 22 shows the workflow for automatic estimation of the UDi. It is worth to mention that all the algorithms and codes for calculation of UDi

Table 3
The results obtained from UDi, KBT, BPDEM and FDEM in probabilistic progressive rock block failure simulation based Experiment 3.

Theta	Steps of simulation	KBT Decision	UDi Kinematic Value	BPDEM Average displacement of formed block (m) in the same steps number	FDEM Average displacement of formed block (m) in the same steps number
45	1	No block created	5.8	0.1	0.00131
	2	No block created	9.97	0.15	0.003
	3	No block created	8.97	0.18	0.005
	4	Stable block	13.1	0.20	0.009
60	1	No block created	5.47	0.0002	0.00085
	2	No block created	9.6	0.06	0.0027
	3	No block created	8.6	0.08	0.0045
	4	Stable block	12.47	0.25	0.0085

based on the virtual sampling of obtained discontinuities from photogrammetry data already exist and is part of the R/Dis/Frag computer package developed by the Director of this project.

The research is conducted within a horizontal tunnel in the sinking level of El Teniente mine's New Level, a property of the National Copper Corporation of Chile (CODELCO-Chile). All the data and essential information forming the basis of this study originates from Digital Terrain Models (DTMs) generated from photographs taken within excavations belonging to a tunnel section of the Sink Level of the Nivel Mina Project, El Teniente Mine (Fig. 23). 52 sequences were mapped and subjected to thorough analysis throughout this span.

5.1. Extracting the fracture sets characteristics

When extracting characteristics from structures mapped in photographic datasets, several key features are systematically identified. The position of each structure is recorded as a georeferenced point in X, Y, and Z coordinates, corresponding to the center of a disc-shaped plane (Cavieres, 2019). The orientation is defined by the dip direction and dip angle relative to these axes. The exposed trace length is measured on the Digital Terrain Model (DTM), providing a close approximation to the true length, even in cases where the structure deviates from a perfectly flat plane. With high-resolution and clean DTMs—combined with prior geological knowledge of the mine—mineral fillings and lithology can also be accurately identified. The thickness of discontinuities is estimated to the nearest centimeter, allowing for precise characterization even of fine structures. Size is inferred from the radius of the disc used during mapping, while roughness is evaluated based on surface morphology, following standardized methodologies. This streamlined and systematic approach ensures accurate mapping and detailed characterization of the mine's structural features. As a result, a total of 3,452 discontinuities were identified across the 52 sections within the study area.

A total of 116 fracture sets were identified across all tunnel sequences. Clustering techniques were employed to automatically identify fracture sets within each tunnel section (Soto et al., 2018). It is important to note that this number represents the cumulative count of fracture sets identified independently at each stage of tunnel advancement. Specifically, 1 fracture set was observed in 6 sequences, 2 sets in 35 sequences, 3 sets in 12 sequences, and 4 sets in 1 sequence, ($6 + 70 + 36 + 4 = 116$). The variation in the number of fracture sets along the tunnel alignment is illustrated in Fig. 24, where each bar represents the number of identified fracture sets within the corresponding tunnel section. Fig. 25 presents a lower hemisphere equal-angle pole plot showing the orientation of all identified fracture sets along the tunnel. The plot reveals a high concentration of fracture sets with azimuths near 0° (North) and steep dips close to 90°. Additionally, a significant number of fractures exhibit moderate dips between 30° and 60°, with azimuths clustering around 90° (East).

5.2. 3D DFN modeling approach

The initial step involves representing the discontinuity network within the rock mass, for which a stochastic Discrete Fracture Network (DFN) approach is recommended. This method enables the generation of

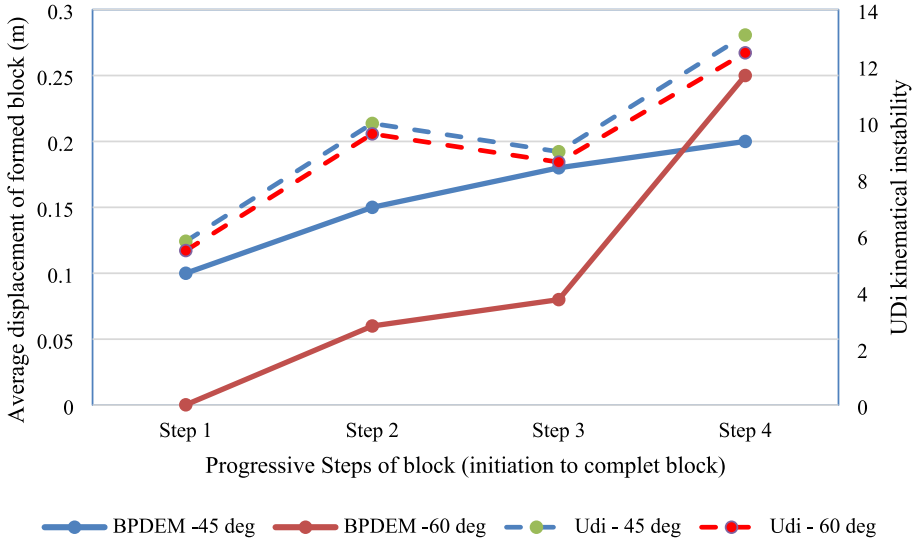


Fig. 19. Application of udi in probabilistic progressive in comparison with the bpdem numerical model-simulation-based experiment 3.

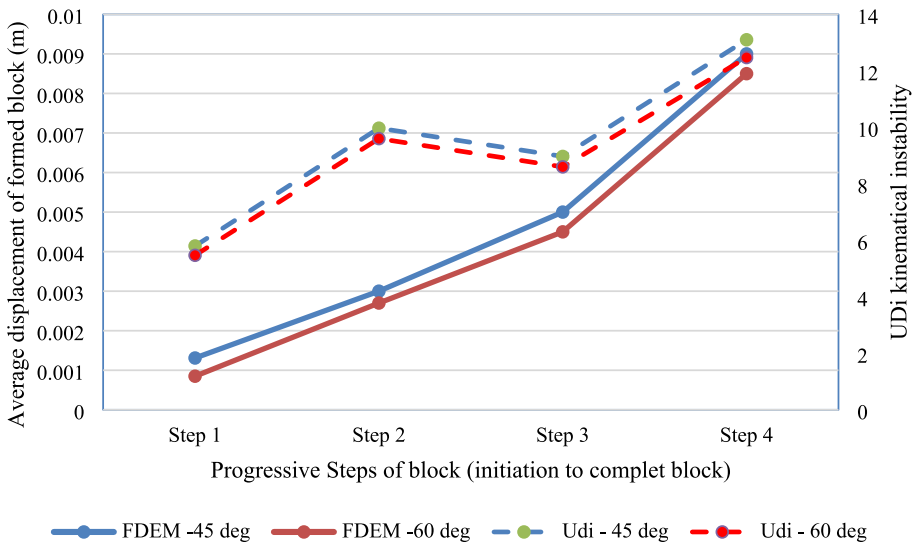


Fig. 20. Application of udi in probabilistic progressive in comparison with the fdem numerical model-simulation-based experiment 3.

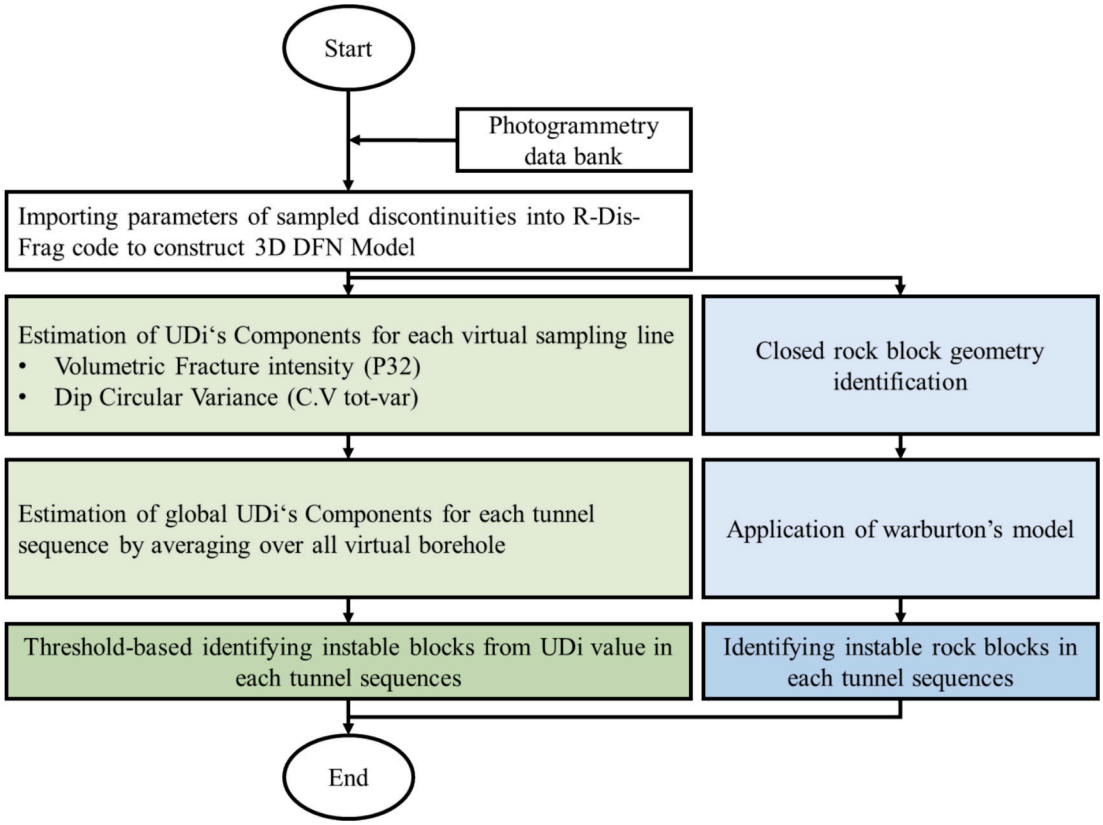


Fig. 21. Flow Chart detailing the comparative methodology to identify high-risk zones for rock block failure, employing the UDi-based approach and Warburton's model.

a realistic fracture network based on field data, which can then be analyzed using a block identification algorithm. Key parameters required for constructing the DFN include the orientation, size, and intensity of discontinuities—each of which is critical for developing a stochastic model that faithfully replicates the fracture behavior and structural complexity of the rock mass (Hekmatnejad et al., 2019).

a) Fracture sets Orientation

In this study, the orientation distribution of fracture (joint) sets is modeled using the Fisher distribution, and the resulting parameters are employed to construct a Discrete Fracture Network (DFN) for each tunnel sequence. Fig. 26 presents a lower hemisphere equal-angle stereographic projection, illustrating both the simulated fracture poles and the field-sampled poles for a representative tunnel sequence. The diagrams highlight a concentration of sub-vertical fracture sets with azimuths near 0° (North), as well as a significant number of fractures with moderate dips between 30° and 60°, predominantly oriented toward 90° (East). The close agreement between the simulated and sampled data indicates that the reconstructed fracture network effectively replicates

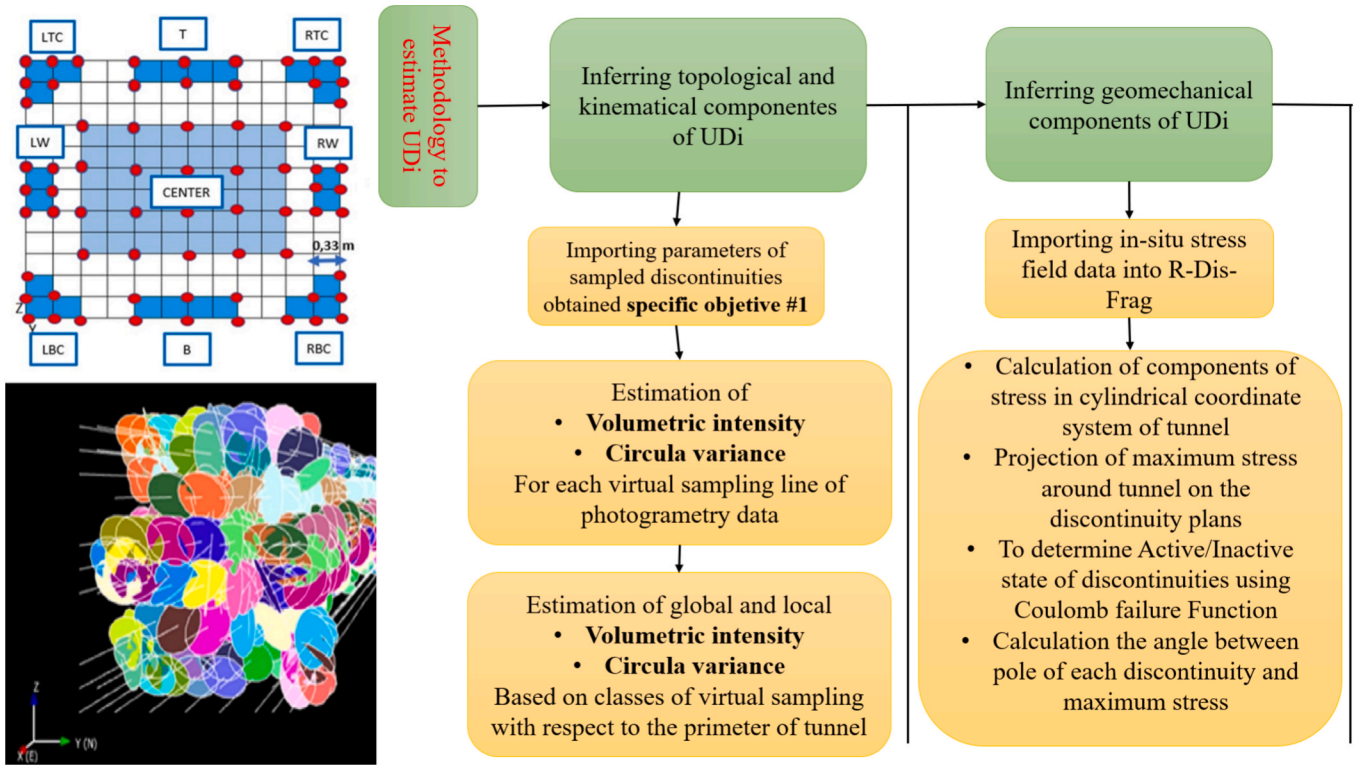
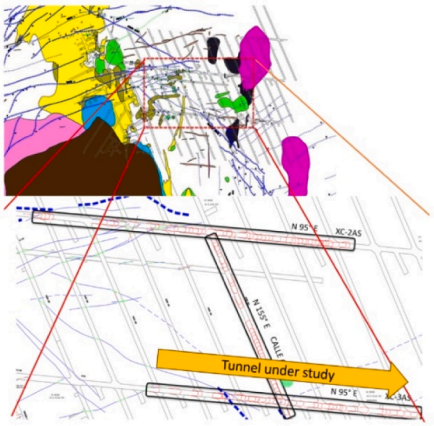
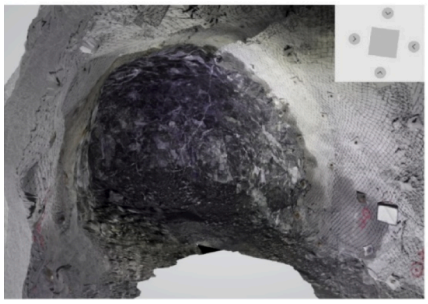


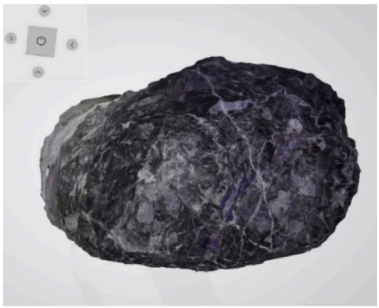
Fig. 22. The designed geometry for sampling and the workflow for automatic estimation of the UDi based on the virtual borehole sampling of discontinuities.



(a)



(b)



(c)

Fig. 23. Case study and dataset representation; a) the study area in the Sinking Level of the El Teniente mine's new Level (Each color presents a geologic unit); b) An example of 3D photogrammetry data (tunnel sequence XC-3AS); c) cropped tunnel face.

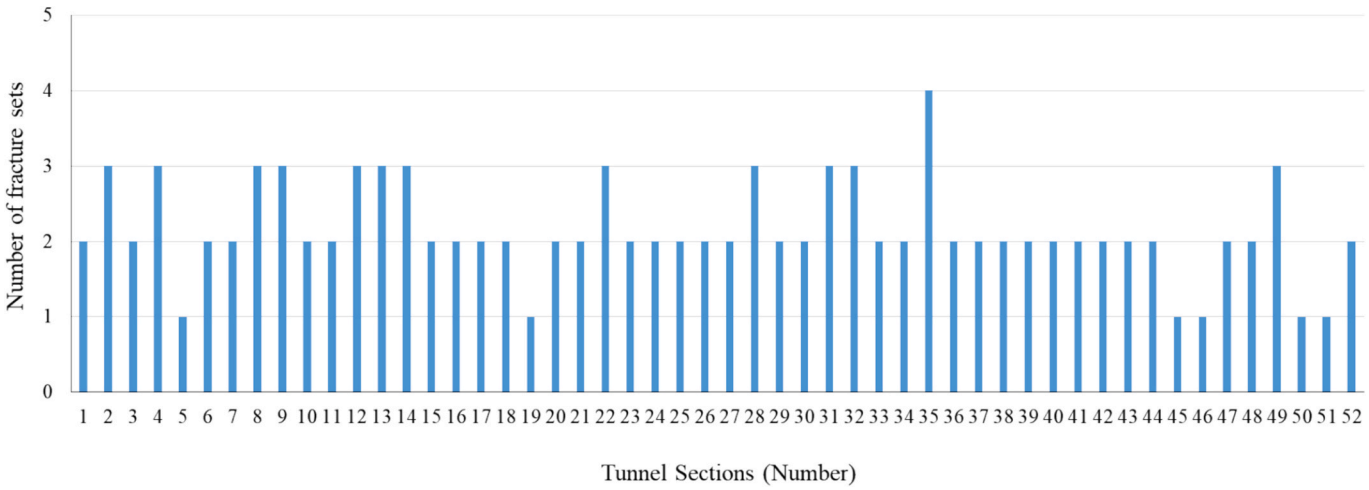


Fig. 24. Bar plot of the number of fracture sets for each sequence of the tunnel.

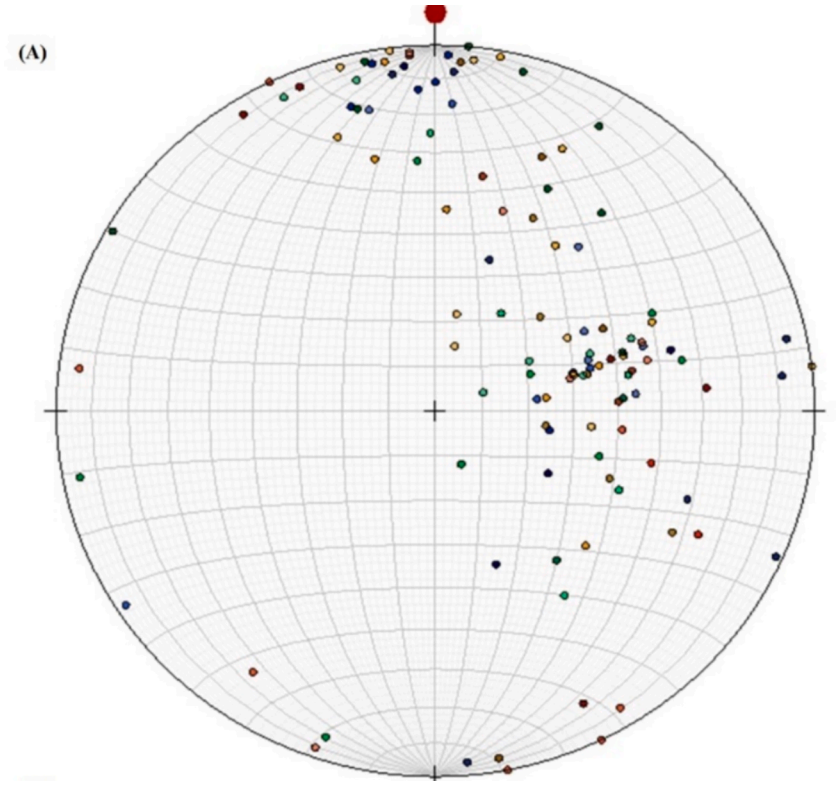


Fig. 25. Fracture sets orientation analysis – Representation of identified fracture sets in the 52 tunnel sequences.

the actual fracture geometry in terms of dip and dip direction.

b) Fracture size distribution

The fracture size distribution and mean fracture diameter were estimated using the method proposed by Hekmatnejad et al. (2018). This approach assumes that fractures are flat circular discs with random orientations and random diameters, spatially distributed according to a homogeneous Poisson process—meaning the average number of fracture centers is constant across the domain. Additionally, it is assumed that fracture diameters and orientations are statistically independent.

The methodology is structured into three main stages, which are described in detail below.

0b.1) Inverting the trace length distribution into the fracture size distribution

The method relates the trace length distribution observed on a planar window (with probability density function $f(l)$) with the fracture size distribution (with probability density function $g(\delta)$ and cumulative density function $G(\delta)$), through the following relationship in **Equation (11)** (Warburton, 1980):

$$f(l) = \frac{1}{\mu}\int_{l}^{+\infty} \frac{g(\delta)d\delta}{\sqrt{\delta^2 - l^2}} \tag{11}$$

where μ stands for the mean fracture diameter. The inversion gives **Equation (12)**:

$$1 - G(\delta) = \frac{2\mu}{\pi}\int_{\delta}^{+\infty} \frac{f(l)dl}{\sqrt{l^2 - \delta^2}} \tag{12}$$

Because μ is unknown, it is convenient to define the rescaled function as

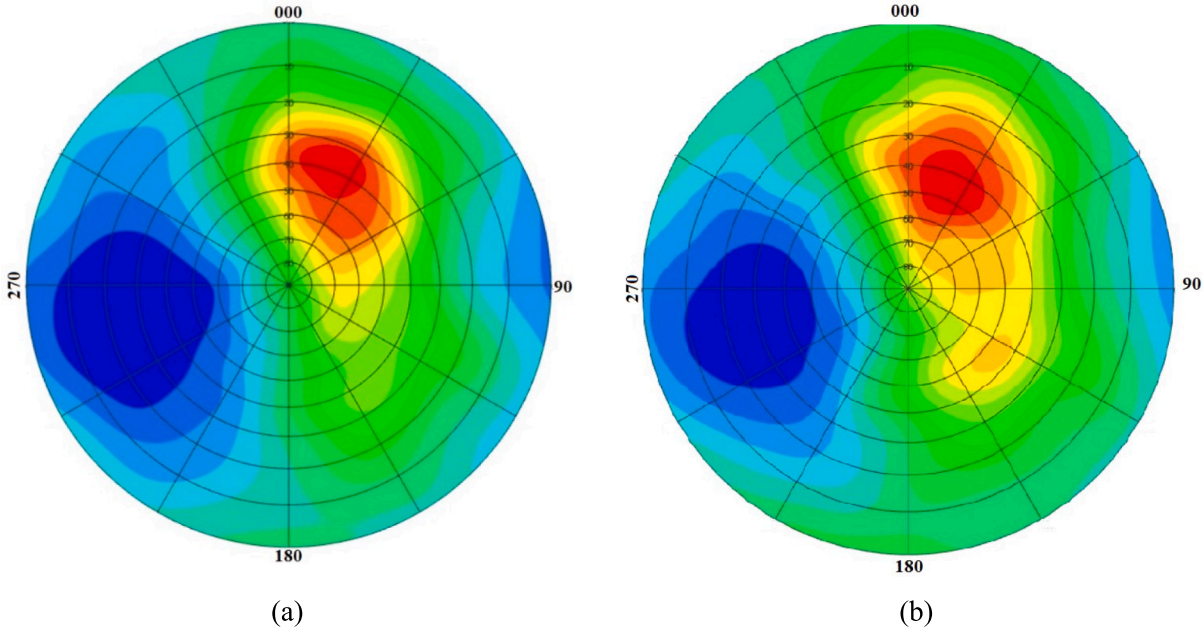


Fig. 26. DFN Analysis- lower hemisphere equal angle contoured pole of a specific tunnel section; a) sampled fractures extracted from photogrammetry for one tunnel sequence; b) simulated fractures network at same sequence.

Equation (13):

$$\varphi(\delta) = \frac{1 - G(\delta)}{\mu} = \frac{2\mu}{\pi} \int_{\delta}^{+\infty} \frac{f(l)dl}{\sqrt{l^2 - \delta^2}}, \tag{13}$$

which can be experimentally determined once an estimation of the trace length distribution $\widehat{f}(l)$ is available, see Hekmatnejad et al. (2018). The estimation is done for a discrete set of diameters $\delta_0, \cdots, \delta_n$, for which one obtains a set of estimated values $\widehat{\varphi}(\delta_0), \cdots, \widehat{\varphi}(\delta_n)$. In practice, one can choose δ_n greater than the maximum trace length so that $\widehat{\varphi}(\delta_n) = \varphi(\delta_n) = 0$.

0b.2) post-processing the estimated function .$\widehat{\varphi}$

The estimated function $\widehat{\varphi}$ so obtained is then corrected for mathematical consistency.

To this end, observe the following:

(1) φ is a nonincreasing function, with values between $\varphi(0) = \frac{1}{\mu}$ and $\varphi(+\infty) = 0$.
(2) The integral of φ is equal to 1; this is because the integral of the complementary cumulative distribution function of a positive random variable is equal to the mean value of this variable (Feller, 1966).

To fulfill the first condition, we first define an upward corrected function $\widehat{\varphi}_1$ in **Equation (14)** and **Equation (15)**:

$$\widehat{\varphi}_1(\delta_i) = \{\widehat{\varphi}(\delta_0) if i = 0 \tag{14}$$

$$\min\{\widehat{\varphi}(\delta_i), \widehat{\varphi}_1(\delta_{i-1}) if i = 1, 2, \cdots, n \tag{15}$$

and a downward corrected function $\widehat{\varphi}_2$ **Equation (16)** and **Equation (17)**:

$$\widehat{\varphi}_2(\delta_i) = \{\widehat{\varphi}(\delta_n) if i = 0 \tag{16}$$

$$max\{\widehat{\varphi}(\delta_i), \widehat{\varphi}_2(\delta_{i+1}) if i = 0, 2, \cdots, n-1 \tag{17}$$

so that $\widehat{\varphi}_1 \leq \widehat{\varphi} \leq \widehat{\varphi}_2$, and both $\widehat{\varphi}_1$ and $\widehat{\varphi}_2$ are non-increasing functions. To fulfill the second condition, we average these two functions in **Equation (18)**:

$$\widetilde{\varphi}(\delta) = a\widehat{\varphi}_1(\delta) + (1 - \alpha)\widehat{\varphi}_2(\delta) \tag{18}$$

with a chosen in $[0, 1]$ in such a way that the final estimated function $\widetilde{\varphi}$ has a unit integral.

Once this consistent estimation $\widetilde{\varphi}$ is obtained, one can derive the mean diameter μ and the cumulative distribution function $G(\delta)$ because $G(0) = 1$ in **Equation (19)** and **Equation (20)**:

$$\widetilde{\mu} = \frac{1}{\widetilde{\varphi}(0)} \tag{19}$$

$$\widetilde{G}(\delta) = 1 - \widetilde{\mu}\widetilde{\varphi}(\delta) \tag{20}$$

0b.3) Determining the uncertainty in the fracture distribution and mean diameter.

The analysis of the fracture size distribution was complemented by a bootstrap study, in which the estimation procedure was repeated 1,000 times on resampled datasets of fracture trace lengths. As a result, 1,000 estimates of both the mean fracture diameter and the cumulative distribution function (CDF) of fracture diameters were obtained, as illustrated in Fig. 27**a and 27b**, respectively. The mean diameter estimated from the original dataset is 2.81 m. In comparison, the bootstrap samples yielded mean diameter estimates ranging from 2.68 m to 2.94 m, with an average value also of 2.81 m, confirming the robustness of the estimation. Fig. 27**c** presents the variation in both the mean size and standard deviation of discontinuity lengths across different tunnel sequences, highlighting spatial heterogeneity in fracture characteristics along the alignment.

c) 3D DFN model.

Given the varying lengths of the tunnel sequences—ranging from 6 to 8 m—it is appropriate to apply the Baecher model (Baecher and Lanney 1978), which assumes that fracture centers are uniformly distributed at the local scale within the domain. For consistency, a fixed generation domain of dimensions 40 m (height) × 40 m (width) × 20 m (depth) was used for all sequences. A tunnel segment corresponding to the specific sequence length is positioned at the center of each domain for further analysis. To enhance the spatial representation of the fracture network, 68 virtual boreholes were strategically distributed along the tunnel's cross-section and perimeter. As illustrated in Fig. 28, this setup

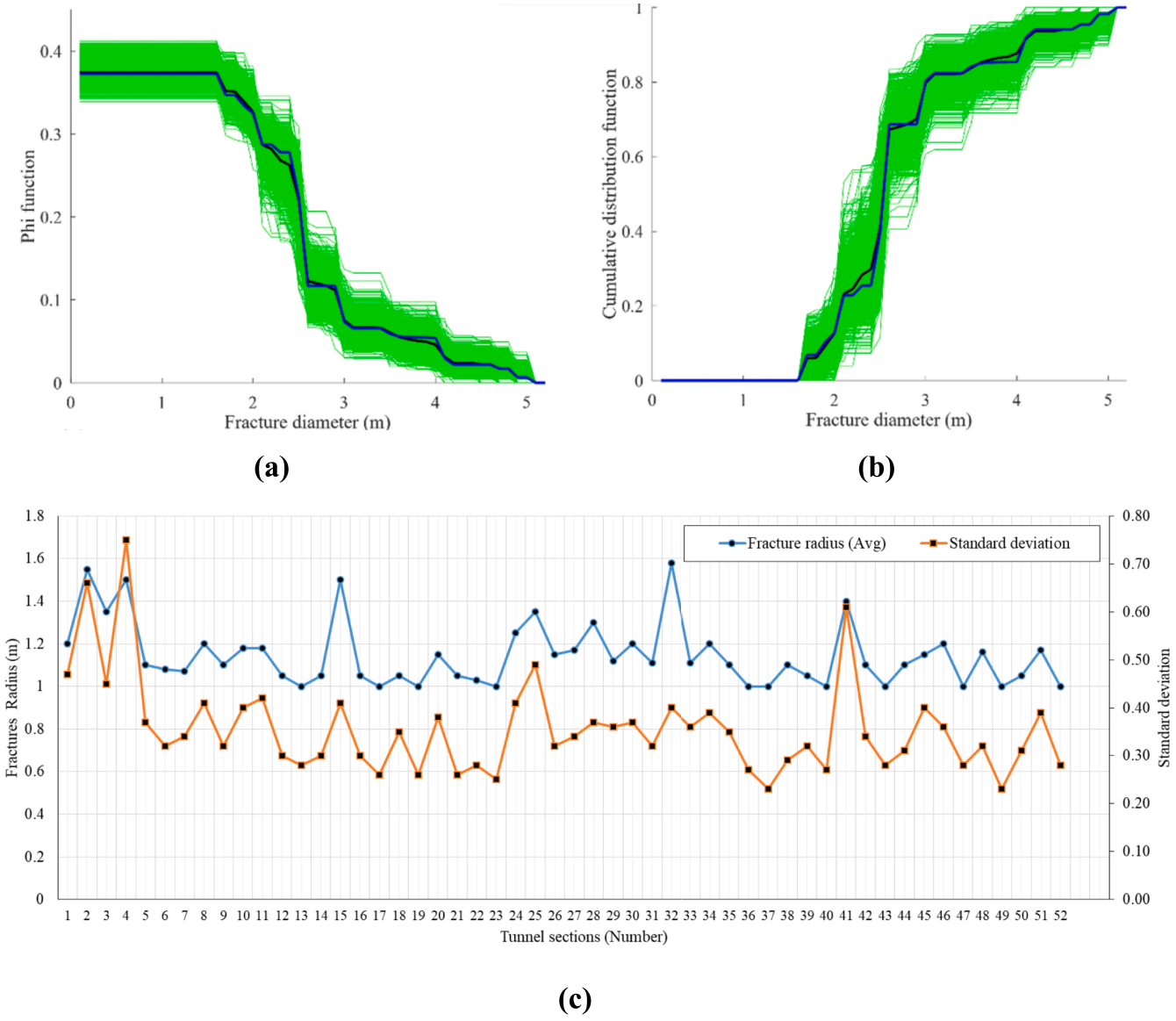


Fig. 27. Fracture sets size distribution; a) Estimation of φ from the original trace length data set (blue), bootstrap samples (green) and average of bootstrap estimates (black). b) Estimation of G from the original trace length data set (blue), bootstrap samples (green) and average of bootstrap estimates (black). c) Mean radius of discontinuities and their standard deviation for each sequence of the tunnel. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

facilitates the estimation of both P32 values (fracture area per unit volume) and the circular variance with respect to the vertical direction, thereby enabling a more detailed structural analysis of each tunnel section. Finally, to account for the inherent uncertainty in the discontinuity system, a Monte Carlo simulation approach was employed. This method generated 100 realizations of the fracture network for each tunnel sequence, resulting in a total of 5,300 Discrete Fracture Network (DFN) models across the entire tunnel alignment.

5.3. Calculating the UDi components

In this study, a standardized approach was adopted by uniformly assigning all discontinuities within the tunnel the same mechanical properties—specifically, a friction angle of 32° and a tensile strength of 17 MPa. This simplification enables a consistent evaluation of failure conditions across the discontinuity network. To estimate the induced stress field around the tunnel, the Kirsch equations were employed, providing a credible first-order approximation of the stress distribution in the surrounding rock mass. This methodological assumption facilitates the direct application of equilibrium equations on the discontinuity planes, thereby streamlining the structural analysis. Table 4 presents the magnitudes and orientations of the principal in-situ stresses around the tunnel, based on field measurements reported by the mining operation.

The volumetric fracture intensity (P32) quantifies the ratio of the total fracture surface area to the volume of rock mass. It integrates information on both the average number of discontinuity centers per unit volume and their corresponding mean diameters. As a scalar metric, P32 does not account for the orientation of discontinuities, focusing solely on their spatial density. In this study, P32 was estimated for each tunnel section using the methodology proposed by Hekmatnejad et al. (2017) and following **Equation (21)**.

$$P32 = \frac{1}{L}\sum_{i=1}^{N}\frac{1}{cos(\alpha_i)} \tag{21}$$

In this study, L represents the scanline length, α is the acute angle between the scanline and the normal vector of the discontinuity, and N

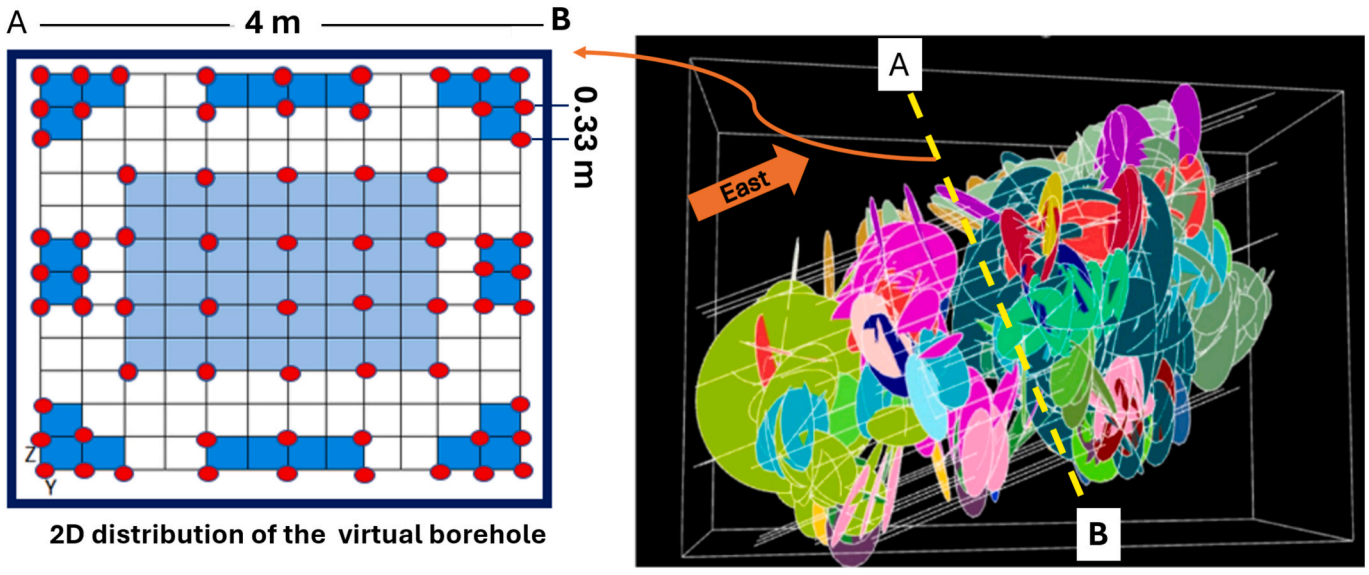


Fig. 28. On the right, a 3D representation the virtual boreholes (white lines) and a realization of fracture networks (colored disk - colors represent id of fractures). On the left, a 2D (A-B cross section) depiction showing the distribution of virtual boreholes at the tunnel face.

Table 4
Measurements of in-situ stress conducted along the length of the tunnel.

Principal stress (MPa)			Azimuth (degree)			Dip (degree)		
Max	Intermediate	Min	Max	Intermediate	Min	Max	Intermediate	Min
59.2	45.3	30.3	135.7	225.0	260.8	5.3	−7.4	80.9

denotes the number of fractures intersecting the scanline. The P32 value for each tunnel section is computed based on the intersections between 68 virtual boreholes—oriented parallel to the tunnel advance—and the discontinuities identified through photogrammetry. The length of each virtual borehole varies according to the dimensions of the corresponding tunnel segment. The average of the 68 individual P32 values is taken as the representative P32 for that tunnel section. Similarly, the dip circular variance for each fracture intersected by the virtual boreholes is calculated using **Equation (5)**, and the average of these values is used to represent the orientation variability (circular variance) of the section.

Fig. 29 illustrates the variation of both parameters along the tunnel alignment, emphasizing the epistemic nature of fracture intensity (P32) and orientation variability, which arise from inherent uncertainty in structural characterization.

5.4. Identifying the high-risk areas for rock block failures using UDi system

To identify high-risk zones for rock block failure, it is necessary to compute the Universal Discontinuity Index (UDi) for each section of the

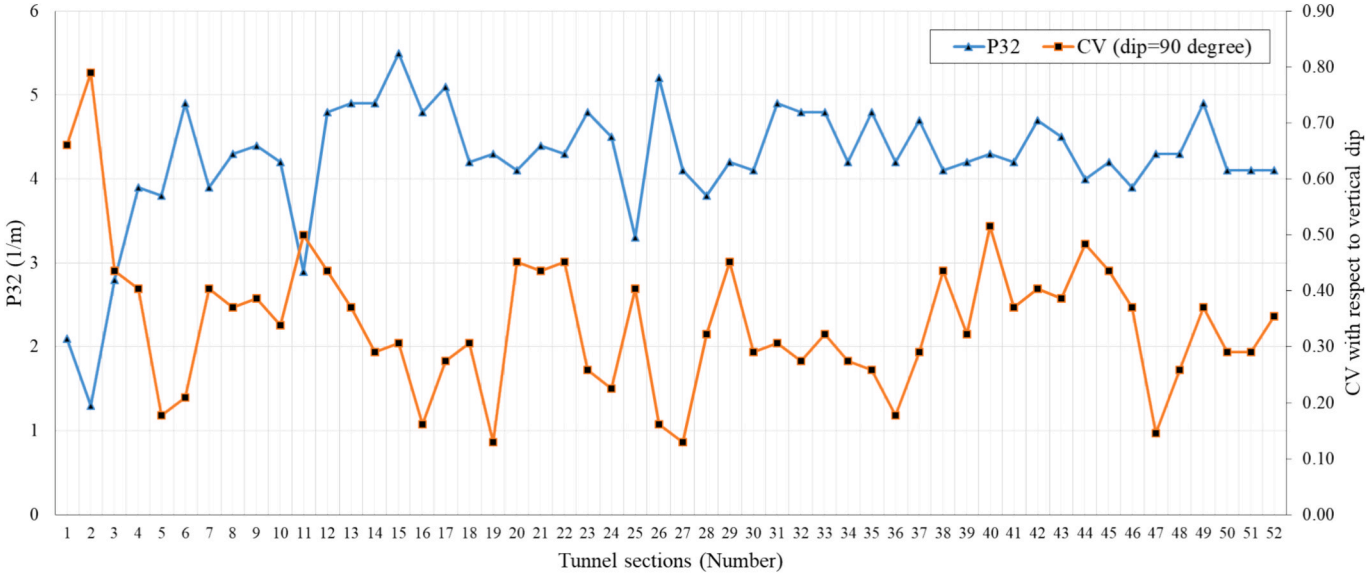


Fig. 29. The estimated values of the volumetric fracture intensity and circular variance with respect to vertical direction for 52 sections of a tunnel.

tunnel under investigation. In this study, the fourth UDi factor is set to zero, based on the assumption that the analysis focuses exclusively on the kinematic feasibility of failure, rather than on mechanical or dynamic conditions.

Tunnel sections are considered kinematically susceptible to failure when two conditions are met: (1) the total circular variance with respect to the vertical direction ($CV._{tot\text{-}ver}$) is equal to or greater than 0.6, and (2) the second UDi factor is non-zero, indicating the presence of active shear discontinuities. These criteria enable the delineation of structurally critical zones within the tunnel alignment.

(Hekmatnejad et al., 2021a) conducted a comprehensive sensitivity analysis to evaluate the influence of P32 and $CV._{tot\text{-}ver}$ (total circular variance with respect to the vertical direction) on the formation of unstable rock blocks, considering a range of average fracture sizes. Their results revealed a significant increase in the likelihood of unstable block formation when P32 exceeds 2.5 and $CV._{tot\text{-}ver}$ surpasses 0.6. Building upon these findings, we propose a predictive criterion for identifying tunnel sections with elevated instability risk. According to this criterion, when $P32 > 2.5$ and $CV._{tot\text{-}ver} > 0.6$, the probability of unstable rock block formation is markedly increased. Moreover, the same study showed that when P32 exceeds 2.5, the dependency on circular variance diminishes, indicating that unstable blocks can still form even if $CV._{tot\text{-}ver}$ is below 0.6. This suggests that in high-intensity fracture environments, P32 becomes the dominant controlling factor in the development of instability.

However, in practical terms, this may not always be the case. This approach considers the overall P32 and $CV._{tot\text{-}var}$ value across the entire tunnel section rather than focusing specifically on the P32 and $CV._{tot\text{-}var}$ of such fractures that contribute to closed rock block formation. Additionally, it is essential to note that this approach does not identify key-light blocks, which refer to rock blocks in direct contact with the free surface of the tunnel. Based on this established criterion, zones with a higher risk of rock block failures can be identified.

According to the formulation of the Universal Discontinuity Index (UDi), the product of P32 and dip circular variance can be interpreted as an indicator of the proportion of fractures that satisfy kinematic failure conditions. Fig. 30 illustrates the variation of the normalized kinematic component of the UDi along the tunnel alignment. Here, normalization refers to dividing UDi values by 2.5, a threshold value identified in previous studies. This normalization is significant, as values below 1 indicate a relatively low likelihood of rock block formation, and thus a reduced risk of instability. Based on this interpretation, the tunnel alignment is categorized into two zones:

- The green zone, where the normalized UDi values are less than 1, indicating a negligible probability of unstable rock block formation.
- The yellow zone, where normalized values are equal to or greater than 1, represents areas where the formation and failure of rock blocks is highly probable.

5.5. Verification of the UDi results

Due to the application of the Warburton method, which requires detailed consideration of rock block geometry in conjunction with each DFN realization, the Key Block Theory (KBT) algorithm was executed 5,300 times—effectively incorporating parameter uncertainties into the block stability analysis. Fig. 31 presents the outcome of this extensive analysis, showing the mean number of unstable rock blocks, alongside the results of the normalized kinematic components of the UDi. The comparison reveals a strong correlation between both approaches, confirming the robustness and reliability of the proposed methodology (see Fig. 32).

Based on these results, the plot is divided into three color-coded zones for risk assessment:

- The green zone represents a low probability of rock block failure, indicating relatively safe conditions—in full agreement with the results presented in Section 4.3 and Fig. 11.
- The yellow zone denotes a moderate risk, suggesting areas that warrant increased caution.
- The red zone identifies areas of high risk, where the likelihood of rock block failure is significant and potentially critical.

5.6. Proposed predictive model for early estimation of unstable rock blocks using UDi

This research proposes a practical methodology for applying the Universal Discontinuity Index (UDi) during the early stages of tunnel excavation to estimate the average number of unstable rock blocks. By employing a Poisson regression model, the study demonstrates the effectiveness of UDi in predicting the formation of removable key blocks with a 95 % confidence interval (see Fig. 33). To validate this approach, the Jackknife resampling technique was utilized. This technique involves splitting the dataset into two subsets without replacement, allowing the model to be trained on one subset and tested on the other. Specifically, data from the first ten tunnel sequences were used to perform a hybrid DFN–Key Block analysis, which served as the basis for calibrating the supervised Poisson regression model. The model uses UDi values derived from virtual boreholes aligned with the tunnel axis to

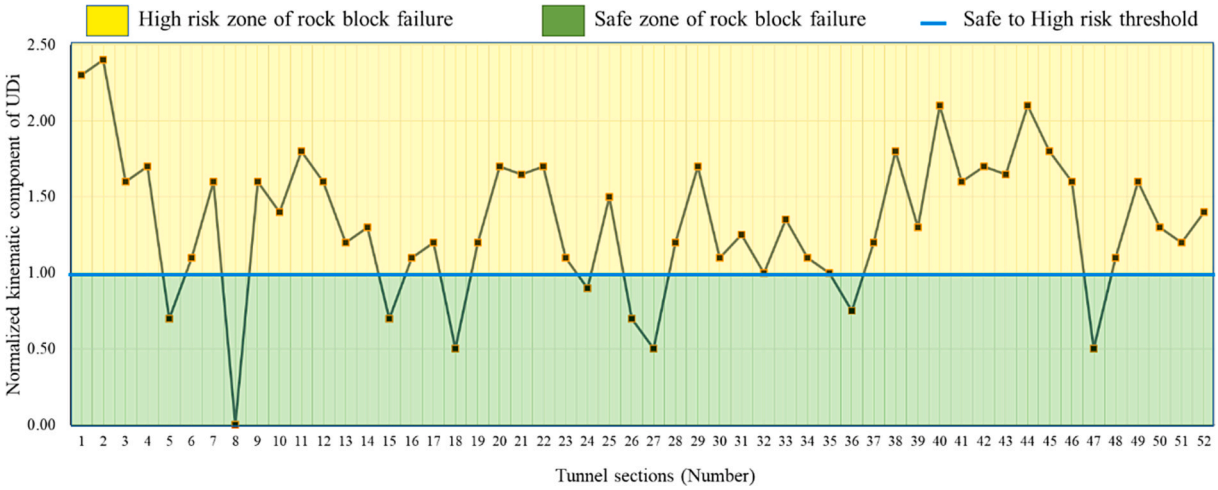


Fig. 30. The vibrational trend of the normalized kinematic components of the UDi along the tunnel sections.

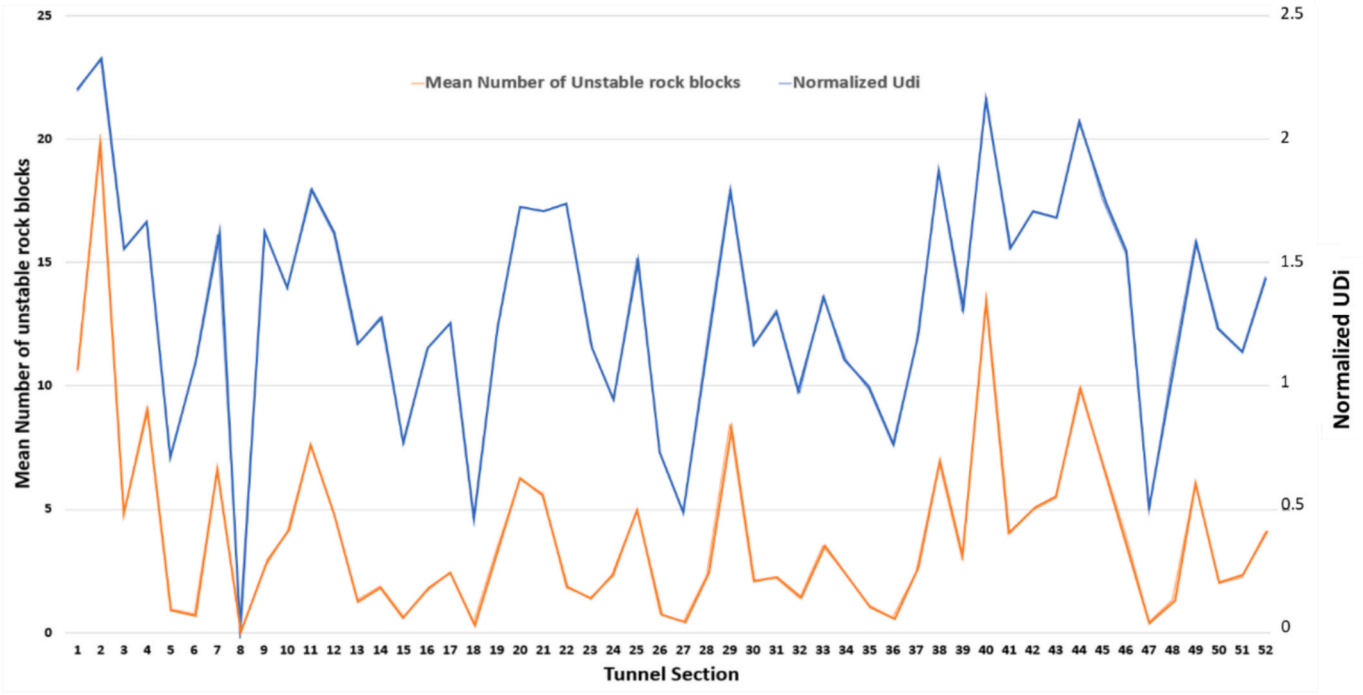


Fig. 31. The outcome of each method for all 52 sequences of a tunnel.

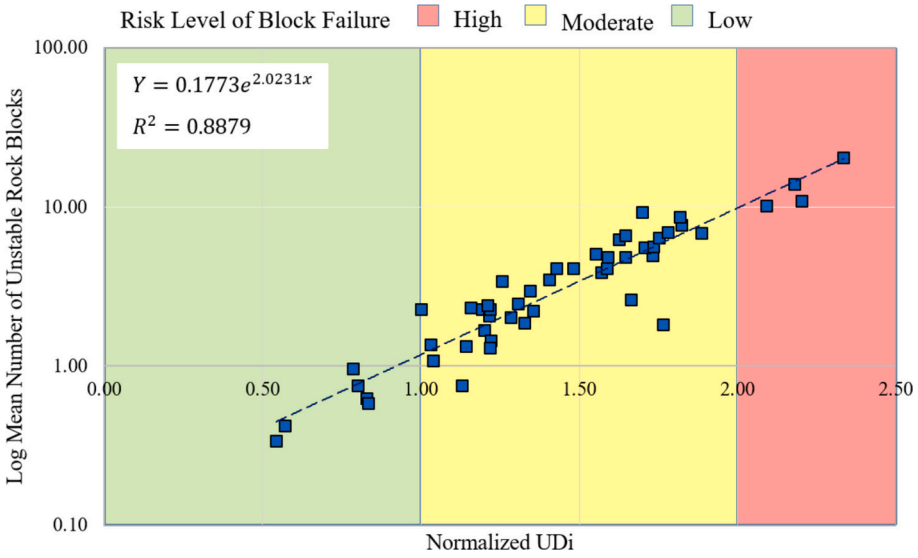
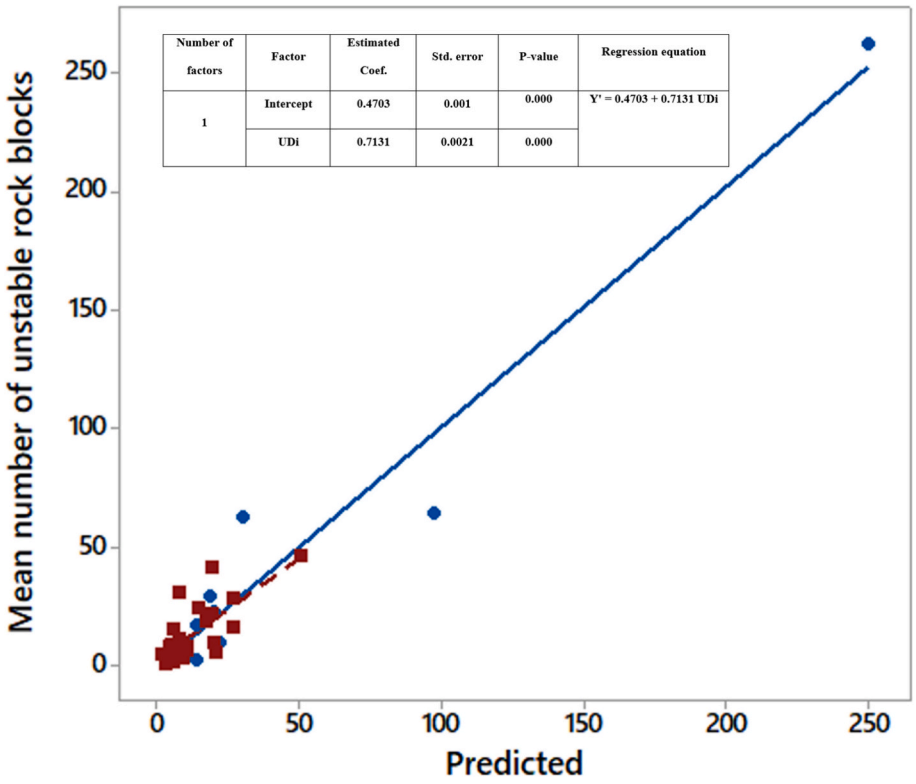


Fig. 32. Enhanced Scatter Plot with Optimized Trend Line Illustrating the Correlation Between the UDi's Normalized Kinematic Components and the Mean Count of Unstable Rock Blocks, Utilizing the Combined DFN-Warburton Approach.

estimate the number of unstable rock blocks. Fig. 33 presents a scatter plot comparing the model's predicted outcomes with the results from the hybrid DFN–KBT analysis, illustrating both the practicality and predictive power of the approach. Importantly, the fitted regression line (solid blue) closely aligns with the actual outcomes from the remaining 42 tunnel sequences (red dashed line), indicating a high level of accuracy. This strong agreement confirms the model's capability to predict the likelihood of removable key block formation based on UDi values, making it a valuable tool for proactive risk assessment and support planning in tunnel excavation projects.

To mitigate potential autocorrelation biases in the Poisson-regression early-warning model—particularly those arising from spatial and temporal clustering of block failures—we implemented several methodological safeguards. First, spatio-temporal stratification was employed by treating each tunnel round as an independent observational unit. Temporal independence was approximated through the use of non-overlapping excavation rounds, while spatial independence was reinforced by excluding adjacent tunnel segments potentially influenced by preceding failures. Second, the model incorporated covariates that capture geomechanical variability across sectors (e.g., fracture density, stress orientation, confinement ratio), which serve to absorb spatial heterogeneity and minimize the confounding effects of unmodeled clustering. Third, model robustness was evaluated via a leave-one-sector-out cross-validation procedure, which confirmed consistent predictive accuracy even when applied to unseen spatial domains, suggesting the model generalizes well and is not overfitted to



Number of factors	Factor	Estimated Coef.	Std. error	P-value	Regression equation
1	Intercept	0.4703	0.001	0.000	Y' = 0.4703 + 0.7131 UDi
	UDi	0.7131	0.0021	0.000	

Fig. 33. Scatter plot of the real number vs. the estimated number of unstable blocks based on a predictive Poisson regression model using Jack-Knife validation technique. The table shows the Maximum likelihood estimation of the coefficients of the multivariate Poisson regression model for normalized UDi.

localized patterns.

To further enhance the treatment of spatial–temporal dependencies, future work will explore the use of spatio-temporal point process models and Bayesian hierarchical frameworks. These approaches are well-suited to explicitly account for the bursty or clustered nature of block failure events. Additionally, recent advances in spatially-aware machine learning—such as Random Forest combined with spatial regression techniques—offer promising alternatives. Notable examples include spatial linear mixed models (Banerjee et al., 2014), RF-GLS (Saha et al., 2023), RF-Residual Kriging (Fayad et al., 2016; Fox et al., 2020), spatial RF (Hengl et al., 2018), and geographical RF (Georganos et al., 2021). These methods may offer improved spatial generalizability and uncertainty quantification, supporting the development of more resilient early-warning systems in geotechnical applications.

5.7. A closed-loop digital twin framework for predictive rock block instability assessment using the universal discontinuity index (UDi)

The predictive framework presented in this study is designed for seamless integration within a closed-loop Digital Twin (DT) architecture, enabling real-engineering time geomechanical hazard forecasting at the pace of daily—or even per-round—tunnel excavation cycles. This DT-enabled early warning system is structured into five interconnected layers that form a fully automated, continuously updating data pipeline:

1 Data Acquisition (Edge Layer).

Following each blast round, high-resolution LiDAR or photogrammetry scans of the newly exposed tunnel face are captured. Simultaneously, field tablets collect and transmit additional inputs such as drilling parameters, geologist assessments via kiosk systems, and microseismic event data. This multimodal data stream ensures a comprehensive representation of the evolving excavation front.

2 Automated Feature Extraction (Processing Layer).

The updated 3D point cloud is processed through the *R-Dis-Frag* engine, which reconstructs the surface mesh and automatically extracts discontinuity traces. These are sampled using pre-registered virtual boreholes embedded within the DT geometry. From this data, the four components of the Universal Discontinuity Index (UDi)—Physical Damage (PD), Effective Stress Intensity Factor (ESIF), Active Fracture Ratio (NAF/NTF), and Kinematic Feasibility (Kine)—are computed in near real-time.

3 Risk Analytics (Twin Core Layer).

The face-averaged UDi values are passed to a calibrated Poisson regression model (as detailed in Section 5), which generates a probabilistic estimate of the number of removable rock blocks expected in the subsequent excavation round, including both the mean and a 95 % confidence interval. A Bayesian updating filter continuously refines model coefficients by assimilating observed block fall outcomes, thus enabling self-learning behavior over time.

4 Decision Support and Visualization (Application Layer).

Risk predictions are visualized on an interactive DT dashboard, built upon an IFC/BIM digital backbone. Each tunnel face is assigned a traffic light classification:

5. **Green** ($\lambda < 1$): Routine condition
6. **Yellow** ($1 \leq \lambda < 3$): Caution
7. **Red** ($\lambda \geq 3$):

Critical Based on the inferred hazard level (λ), the system auto-generates targeted support recommendations (e.g., additional pattern bolts, mesh reinforcement), which are displayed alongside the 3D mesh model for clear, actionable guidance.

8 Action and Feedback (Physical Layer).

The shift supervisor either confirms or modifies the recommended support actions. This decision, once executed, is logged and re-integrated into the DT pipeline—closing the feedback loop and updating the UDi computation and model predictions accordingly. If λ exceeds a predefined alert threshold, the system automatically dispatches SMS/email notifications to the geotechnical engineering team, providing true early-warning capability rather than reactive post-event analysis.

By explicitly mapping the full chain of operations—**Acquire** →

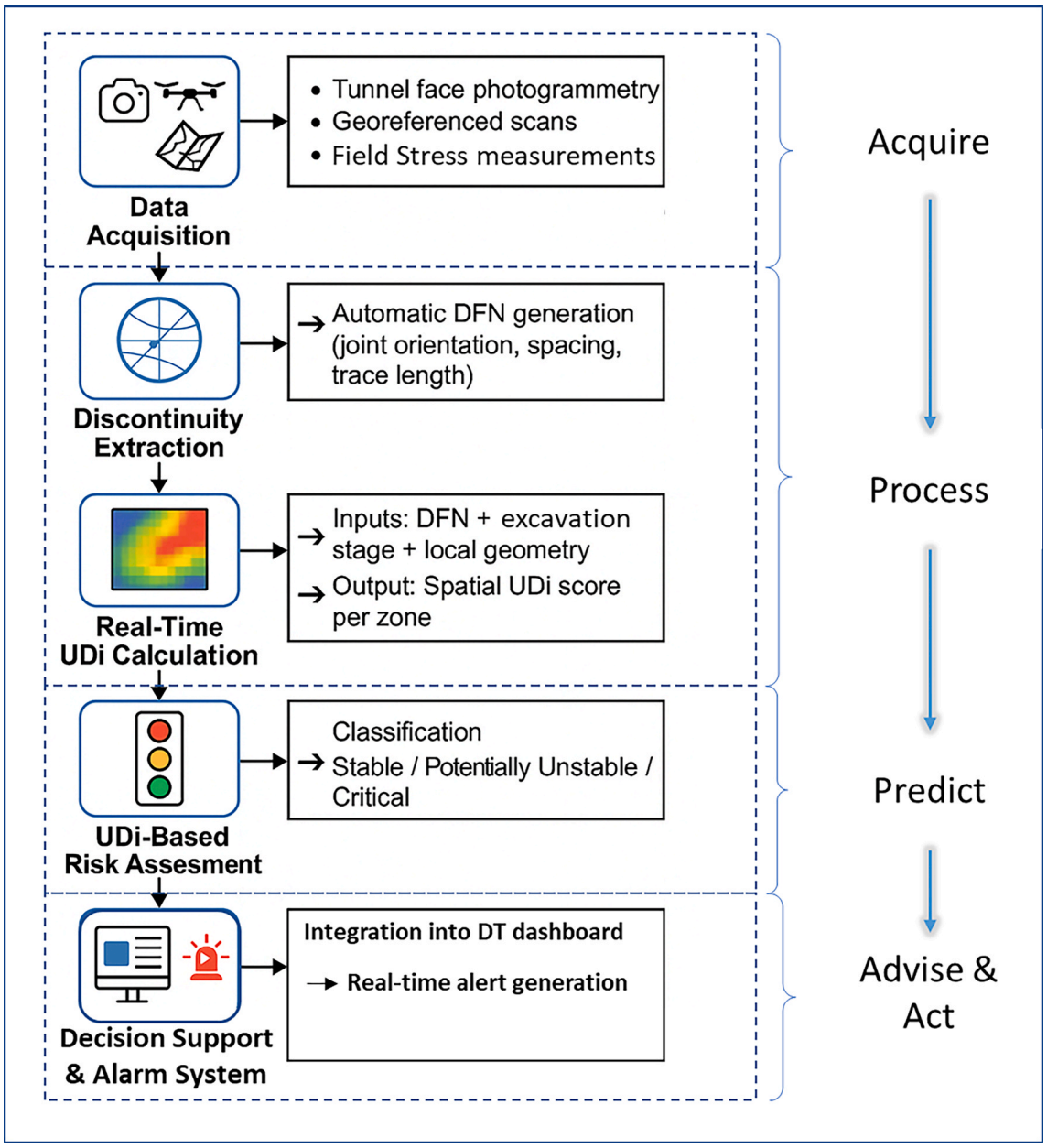


Fig. 34. Schematic workflow of the real-time Digital-Twin system for tunnel instability forecasting based on the Universal Discontinuity Index (UDi).

Process → Predict → Advise → Act—into a unified, automated DT workflow (as shown in Fig. 34), this system transforms UDi-based rock mass characterization into a predictive, adaptive, and intelligent tool for proactive ground support planning and hazard mitigation.

6. Conclusions

This research delivers a significant advancement in underground geomechanical risk assessment by establishing the Universal Discontinuity Index (UDi) as a robust, scalable, and real-time framework for evaluating probabilistic rock block instability. UDi is providing a quantitative, dynamic, and probabilistic alternative to traditional deterministic models. The key conclusions drawn from this study are as follows:

Superior Predictive Capability Across Scales: UDi consistently outperforms conventional approaches, including the Key Block Theory (KBT), by capturing both static and progressive failure mechanisms in jointed rock masses. In controlled simulation experiments, UDi accurately replicated the temporal evolution of instability observed in BPDEM and FDEM simulations for both simple and complex wedge geometries. Field validation at El Teniente mine further confirmed UDi's predictive strength, with the normalized kinematic component achieving a high correlation ($R2 = 0.93$) with the mean number of unstable blocks derived from 5,300 DFN–KBT realizations.

Probabilistic, Multifactorial, and Scalable Framework: The UDi formulation synthesizes four core, physically grounded components: Physical Damage (PD), Effective Stress Intensity Factor (ESIF), Active Fracture Ratio (NAF/NTF), and the Kinematic Factor (Kine). This integration enables UDi to probabilistically assess damage and instability, scalable from laboratory samples to full tunnel and cavern complexes. Its computational efficiency—delivering updates within milliseconds—makes it suitable for real-time risk assessment and decision support during active excavation.

Complete System View and Deep Model Integration: UDi embodies a complete system perspective inspired by enterprise system architectures. It deeply integrates structural, mechanical, and kinematic parameters into a unified risk metric. This holistic approach supports cross-disciplinary understanding and aligns with modern engineering integration standards, bridging geology, geomechanics, and mine planning. The semantic clarity of UDi facilitates communication among stakeholders, promoting interoperability across different domains within the mining and tunneling sectors.

Fracture Propagation-Informed Refinement: A key innovation of UDi is its ability to account for evolving fracture networks and the transitional geometry of partially formed blocks—features overlooked by classical models. This capability enables continuous refinement of instability assessments as excavation progresses, supporting adaptive hazard mapping and proactive intervention strategies.

Field-Proven Early-Warning System: The Poisson regression model integrated with UDi translates the index into a predictive tool capable of estimating the number of unstable blocks expected per tunnel round. Validated through jackknife resampling, this approach demonstrated 95 % prediction confidence, offering practical utility for early warning and support design optimization in real tunnel operations.

System Integration and Digital Twin Readiness: UDi's architecture is inherently compatible with Digital Twin workflows. It functions as a semantic integration layer that fuses heterogeneous data sources—such as photogrammetry, LiDAR, and microseismic monitoring—into a coherent and quantitative risk assessment platform. The UDi framework thus operationalizes DT principles, enabling continuous model updating, real-time risk visualization, and intelligent decision support in tunneling projects.

Validated Reliability Across Diverse Geological Conditions: UDi has demonstrated robust reliability and reproducibility across laboratory, numerical, and field studies under varying geological conditions. Applications include tunnel damage rating (Hekmatnejad et al., 2022), spatial modeling of rock strength heterogeneity (Hekmatnejad et al., 2023), structural controls on drill-and-blast overbreak (Cortéset al, 2025), and probabilistic hazard mapping.

Closed-Loop Geotechnical Data Cycle: The study operationalizes a closed-loop data cycle within the DT context: data acquisition (photogrammetry, LiDAR) → automated feature extraction (via R-Dis-Frag) → UDi computation → probabilistic forecasting (Poisson model) → real-time operational feedback (DT dashboard). This loop allows for on-the-fly updates to risk assessments and supports adaptive excavation management without the need for time-consuming remodelling.

Bridging Gaps Between Deterministic, Numerical, and Predictive Models: UDi effectively bridges the gap between deterministic kinematic models, such as Key Block Theory (KBT), high-fidelity numerical simulations (BPDEM and FDEM), and real-time predictive analytics. While KBT provides only a static and geometry-based assessment, and numerical methods, though accurate, are computationally intensive and time-consuming, UDi offers a computationally efficient alternative. It captures progressive failure mechanisms with predictive accuracy comparable to numerical models while enabling rapid, scalable, and probabilistic risk assessments suitable for real-time decision-making in tunnel operations.

UDi's comprehensive, system-integrated, and predictive approach enhances our capacity to understand, anticipate, and mitigate structurally controlled failures in complex geological settings. UDi's scalability, adaptability, and computational efficiency make it a cornerstone for safer, smarter, and more resilient underground infrastructure development in both mining and civil engineering applications.

CRediT authorship contribution statement

Amin Hekmatnejad: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Mahdi Bajolvand:** Writing – review & editing, Visualization, Validation, Software, Methodology. **Pengzhi Pan:** Writing – review & editing, Validation, Methodology, Investigation. **Xavier Emery:** Writing – review & editing, Validation, Methodology, Investigation. **Alvaro Pena:** Validation, Methodology, Investigation. **Jorge Prado:** Validation, Methodology, Investigation. **Abbas Taheri:** Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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